
Robust Filter Attention: Self-Attention as Precision-Weighted State Estimation

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Abstract

We introduce Robust Filter Attention (RFA), a formulation of self-attention as a robust state estimator. Each token is treated as a noisy observation of a latent trajectory governed by a linear stochastic differential equation (SDE), and attention weights are determined by consistency under this model rather than static feature similarity. Under isotropic noise and decay assumptions, RFA matches the computational complexity of standard attention. On language modeling benchmarks, RFA achieves lower perplexity than RoPE within the training window while remaining stable under zero-shot extrapolation to longer contexts. The framework also provides a dynamical interpretation of standard positional mechanisms, connecting rotational embeddings and recency biases to transport and uncertainty propagation induced by stochastic dynamics.

1. Introduction

Self-attention has become the dominant paradigm for sequence modeling due to its parallelism and scalability (Vaswani et al., 2017). Unlike recurrent architectures (Elman, 1990), however, it does not explicitly propagate latent states through shared temporal dynamics — each token attends independently to all others, with no constraint that states evolve consistently over time. Temporal structure is therefore encoded through positional mechanisms rather than state evolution.

We introduce Robust Filter Attention (RFA), which grounds such structure in an explicit stochastic dynamical prior: past tokens are propagated to the query position under learned linear dynamics, and attention weights are determined by their predicted reliability under this model, coupling transport and uncertainty propagation.

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In RFA, each token is modeled as a noisy observation of a latent trajectory. The query token serves as a reference observation of the current state, while past keys are propagated to the query position under learned dynamics, each yielding a prediction of the latent state with a precision given by the Differential Lyapunov Equation (DLE). Attention weights are then computed from prediction errors measured under these precisions, so that tokens are weighted according to their consistency with the dynamical model. Unlike recursive filters such as the Kalman filter, RFA performs estimation independently at each query position, yielding a parallel batch estimator compatible with attention-style computation. The SDE serves only as a prior governing the propagation of information and uncertainty, while attention retains direct access to historical observations rather than compressing them into a recurrent state.

Depending on the learned balance between process and measurement uncertainty, different heads may specialize into distinct temporal filtering behaviors, ranging from short-range recency-biased filtering to stable long-range integration.

Common positional encodings can be understood as imposing implicit dynamical models on how information evolves across tokens. RoPE (Su et al., 2024) encodes transport through norm-preserving rotations but assigns no explicit reliability to tokens as a function of distance. ALiBi (Press et al., 2022) imposes a distance-based reliability penalty but carries no corresponding transport operator. In RFA, transport and reliability are not independent design choices but consequences of a shared dynamical model. RFA under noiseless, decay-free dynamics recovers RoPE, while pure diffusion with no decay yields a logarithmic distance penalty analogous to ALiBi.

Finally, we introduce Spectrally-Coupled RFA (SC-RFA). Standard rotational encodings such as RoPE apply no decay, so high-frequency components persist indefinitely. As a result, tokens far apart in the sequence can appear positionally similar to nearby ones, degrading long-context discrimination. SC-RFA addresses this by partitioning the frequency spectrum across attention heads and coupling each head’s decay rate to its maximum frequency, so that high-frequency heads act as short-range filters while low-frequency heads behave as stable long-range integrators.

Our contributions are as follows:

- (i) A derivation of self-attention as a tractable robust batch estimator for a linear SDE.
- (ii) A scalable isotropic formulation with analytic uncertainty modeling at standard attention cost.
- (iii) A recovery of RoPE and ALiBi as limiting cases, corresponding respectively to noiseless, zero-decay dynamics and Brownian diffusion.
- (iv) A spectrally-coupled decay prior enabling multi-resolution temporal filtering across heads, with principled suppression of long-range phase interference.

2. Related Work

2.1. Probabilistic and Kernel Views of Attention

The Transformer architecture (Vaswani et al., 2017) computes attention scores via a scaled dot-product between queries and keys. While originally motivated by its efficiency and ability to capture long-range dependencies, a growing body of work has sought to interpret these weights as probabilities derived from latent statistical models.

Probabilistic Transformers (Gabbur et al., 2021) show that dot-product attention arises as a constrained limit of MAP inference in a Gaussian mixture model. The Bayesian Attention Mechanism (BAM) (Bianchessi et al., 2026) treats positional embeddings as explicit priors over token indices, while the Correlated Gaussian Process Transformer (CGPT) (Bui et al., 2024) interprets asymmetric projections through correlated Gaussian process inference. These approaches introduce probabilistic structure, but rely on static feature-space similarity or fixed prior covariances.

Other work has interpreted attention through the lens of kernel regression, identifying softmax attention with the classical Nadaraya-Watson estimator (Tsai et al., 2019). Subsequent work has explored alternative kernels and weighting schemes to improve robustness, expressivity, or statistical efficiency (Han et al., 2023; Nielsen et al., 2024; Liu et al., 2020). These approaches treat attention as a kernel smoother whose behavior is determined by the choice of kernel. In contrast, RFA begins from a latent dynamical state estimation problem. The resulting attention kernel is not specified a priori, but emerges from uncertainty propagation and consistency under a stochastic dynamical model.

2.2. Filtering, Continuous Dynamics, and SSMs

Continuous-time sequence models often parameterize latent dynamics using Neural Ordinary Differential Equations (Neural ODEs) (Chen et al., 2018) and their stochastic extensions, Neural SDEs (Li et al., 2020; Shen & Cheng, 2025), which learn drift and diffusion functions from data.

Several architectures integrate attention with continuous dynamics to handle irregular sampling or time-dependent relevance, including Continuous-Time Attention (Chien & Chen, 2021), Attentive Neural Processes (Kim et al., 2019), and ACE-NODE (Jhin et al., 2021). Self-Modulating Attention (SMA) (Chen et al., 2021) adjusts attention weights as a function of temporal distance.

Other work integrates neural networks with classical filtering frameworks by learning components of the Kalman filter, such as gains, noise models, or update rules (Jahan-shahi & Zhu, 2026; Revach et al., 2022; Liu et al., 2023; Cohen & Klein, 2025; Shen et al., 2025). Recent work has also shown that Transformer architectures are expressive enough to approximate a Kalman filter (Goel & Bartlett, 2024). RFA approaches the connection from the opposite direction: rather than showing that attention can represent a filter, we formulate self-attention itself as a batch state estimator.

State space models (SSMs) provide another approach to sequence modeling by assuming linear time-invariant (LTI) dynamics and converting recurrence into convolution. Frameworks such as HiPPO (Gu et al., 2020) and S4 (Gu et al., 2022) achieve efficiency by restricting the dynamics to structured forms (e.g., diagonalizable or diagonal-plus-low-rank), reducing the cost of state updates from $\mathcal{O}(d^2)$ to $\mathcal{O}(d)$ and enabling fast convolutional implementations. Recent work shows that causal linear attention can be viewed as a special case of LTI convolution (Dao & Gu, 2024).

Concurrent with our work, Kalman Linear Attention (KLA) (Shaj et al., 2026) develops a probabilistic filtering interpretation for linear attention and state-space sequence models using parallelized information-form Kalman updates. Whereas KLA and recurrent SSMs propagate a compressed hidden state through recursive updates, RFA instead formulates self-attention as a parallel precision-weighted batch estimation problem, where pairwise uncertainties are analytically derived from stochastic dynamics rather than recursively propagated through posterior state updates, while preserving explicit content-based token interactions through attention.

2.3. Positional Encodings

Modeling relative temporal structure in Transformers has been approached with several geometric methods. RoPE (Su et al., 2024) encodes relative position through deterministic complex rotations of queries and keys, but introduces no explicit notion of decay or uncertainty. ALiBi (Press et al., 2022) applies a linear distance-based bias to attention logits, improving length extrapolation by suppressing distant interactions. xPos (Sun et al., 2023b;a) generalizes RoPE by combining rotations with dimension-wise decay to stabilize long-range behavior.

These methods impose useful geometric or monotonic structure, but are not derived from an explicit model of latent state evolution and measurement. As a result, transport and reliability are treated as separate design choices. Each method can be interpreted as corresponding to an implicit dynamical assumption. RoPE implements deterministic phase transport without uncertainty accumulation, corresponding to a noiseless LTI system with zero decay. ALiBi’s linear distance penalty approximates RFA’s DLE-derived bias under pure Brownian diffusion, giving it an implicit noise model but no transport operator. xPos introduces decay without deriving it from a covariance model, leaving transport and reliability decoupled.

RFA derives both the transport operator and the precision weighting from the same underlying SDE, so that these are coupled by construction rather than introduced independently. This also implies a rotate–aggregate–counter-rotate structure on the value stream, required for aggregation to correspond to fusion of state estimates in a shared temporal frame. While methods such as RoPER (Harik & Jayasiri, 2022) have applied value rotations previously, RFA derives this structure as a necessity of the dynamical model rather than a geometric heuristic.

Methods such as YaRN (Peng et al., 2024) address RoPE’s extrapolation failure by compressing position indices at inference, mitigating out-of-distribution phase rotations post-hoc while introducing a train-inference mismatch. SC-RFA instead prevents the failure by coupling each head’s decay rate to its maximum frequency during training, so that high-frequency modes attenuate before accumulating spurious phase matches. Selective RoPE (Movahedi et al., 2026) also combines rotation with decay via a spectral leakage analogy, but lacks a precision kernel, value rotation structure, and principled frequency-decay coupling — components that in RFA are necessary consequences of the dynamical model rather than design choices.

3. Methods

RFA models each token as a noisy observation of a latent trajectory evolving under a linear SDE. Keys are transported to the query position under the dynamics model, while the Differential Lyapunov Equation provides an analytic estimate of the uncertainty accumulated during transport. Attention weights are then determined by prediction consistency under this uncertainty model, yielding a robust precision-weighted estimator. Under isotropic noise and decay assumptions, the uncertainty model reduces to a scalar function of temporal lag, recovering standard $\mathcal{O}(N^2d)$ attention complexity.

Appendix A provides additional details on the derivation of the estimator, and Appendix B on its formulation as an efficient attention mechanism.

3.1. Attention as Batch State Estimation

Generative model. We model each token embedding $z_i \in \mathbb{R}^d$ as a noisy observation of a latent state evolving under a linear time-invariant SDE:

$$\begin{aligned} d\mathbf{x}(t) &= \mathbf{A}\mathbf{x}(t)dt + \mathbf{G}d\mathbf{w}(t), \\ z_i &= \mathbf{C}\mathbf{x}(t_i) + \mathbf{v}_i, \quad \mathbf{v}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \end{aligned} \quad (1)$$

where $\mathbf{w}(t)$ is a standard Wiener process, $\mathbf{v}_i$ is Gaussian measurement noise with covariance $\mathbf{R}$, and $\mathbf{Q} := \mathbf{G}\mathbf{G}^\top$ is the process noise covariance. We assume invertible $\mathbf{C}$; non-invertible maps correspond to partially observed systems, which we do not consider here.

The SDE functions as a learned dynamical prior rather than an exact model of sequence evolution. Its role is to provide transport and uncertainty estimates that capture recurring temporal structure. We adopt an LTI structure because it admits closed-form propagation of both state means and covariances.

Estimation setup. Our goal is to estimate the latent state $\mathbf{x}(t_i)$ from past observations $\{z_j\}_{j \leq i}$. This estimation problem is solved independently for each query position, yielding a parallel batch estimator rather than a recursive filter, since attention requires comparing the query against individual transported observations rather than a single compressed state. Each past embedding $\hat{\mathbf{x}}_j = \mathbf{C}^{-1}z_j$ is propagated forward via the state transition matrix to form a prediction of the current latent state $\mathbf{x}(t_i)$,

$$\hat{\mathbf{x}}_{ij} = e^{\mathbf{A}\Delta t_{ij}} \hat{\mathbf{x}}_j,$$

where $\Delta t_{ij} = t_i - t_j$. Under the SDE, the prediction error of the transported observation is

$$\mathbf{r}_{ij} := \mathbf{x}(t_i) - \hat{\mathbf{x}}_{ij}, \quad \mathbf{r}_{ij} \sim \mathcal{N}(\mathbf{0}, \Sigma_{ij}),$$

where the covariance is

$$\Sigma_{ij} = \mathbf{V}(\Delta t_{ij}) + e^{\mathbf{A}\Delta t_{ij}} \mathbf{R}_c e^{\mathbf{A}^\top \Delta t_{ij}} + \sigma_0^2 \mathbf{I}.$$

where $\mathbf{R}_c = \mathbf{C}^{-1} \mathbf{R} \mathbf{C}^{-\top}$, and $\mathbf{V}(\Delta t)$ is the solution of the Differential Lyapunov Equation (DLE):

$$\dot{\mathbf{V}}(s) = \mathbf{A}\mathbf{V}(s) + \mathbf{V}(s)\mathbf{A}^\top + \mathbf{Q}, \quad \mathbf{V}(0) = \mathbf{0}, \quad (2)$$

which describes the accumulation of process uncertainty. We additionally include an isotropic noise floor $\sigma_0^2 \mathbf{I}$ to regularize the covariance and account for irreducible uncertainty.

Latent state estimation. The optimal estimator for $\mathbf{x}(t_i)$ under the full joint model requires inverting a dense cross-token covariance matrix. To obtain a tractable attention-style estimator, we neglect cross-token correlations induced by shared process noise while retaining the marginal covariance of each transported observation. We then estimate the

latent state by minimizing a sum of squared Mahalanobis residuals:

$$\bar{\mathbf{x}}_i = \arg \min_{\mathbf{x}} \sum_{j \leq i} (\mathbf{x} - \hat{\mathbf{x}}_{ij})^\top \mathbf{P}_{ij} (\mathbf{x} - \hat{\mathbf{x}}_{ij}),$$

where $\mathbf{P}_{ij} := \Sigma_{ij}^{-1}$. This yields the closed-form estimator:

$$\bar{\mathbf{x}}_i = \left(\sum_{j \leq i} \mathbf{P}_{ij} \right)^{-1} \sum_{j \leq i} \mathbf{P}_{ij} \hat{\mathbf{x}}_{ij}. \quad (3)$$

Shared process noise induces cross-token correlations that are neglected by this factorized approximation. The resulting estimator becomes exact in the limit of zero process noise, where the joint likelihood factorizes and Eq. (3) coincides with the MLE of the full model. Away from this limit, the approximation preserves the exact marginal covariance of each transported observation, so the resulting precision weights remain individually consistent with the uncertainty predicted by the SDE.

Observable residual. The ideal estimator depends on the latent residual $\mathbf{r}_{ij} = \mathbf{x}(t_i) - \hat{\mathbf{x}}_{ij}$, but this quantity is not directly observable because the latent state $\mathbf{x}(t_i)$ is unknown. We therefore replace it with the observable residual

$$\tilde{\mathbf{r}}_{ij} := \hat{\mathbf{x}}_i - \hat{\mathbf{x}}_{ij}.$$

We interpret $\hat{\mathbf{x}}_i = \mathbf{C}^{-1} \mathbf{z}_i$ as the query and $\hat{\mathbf{x}}_{ij} = e^{\mathbf{A} \Delta t_{ij}} \hat{\mathbf{x}}_j$ as the transported key. Replacing the latent state with its noisy observation inflates the residual covariance by the query-side measurement noise:

$$\tilde{\mathbf{r}}_{ij} \sim \mathcal{N}(\mathbf{0}, \tilde{\Sigma}_{ij}), \quad \tilde{\Sigma}_{ij} = \underbrace{\Sigma_{ij}}_{\text{key-side}} + \underbrace{\mathbf{R}_c}_{\text{query-side}}.$$

Letting $\tilde{\mathbf{P}}_{ij} := \tilde{\Sigma}_{ij}^{-1}$, the similarity between query $\hat{\mathbf{x}}_i$ and key $\hat{\mathbf{x}}_j$ is measured by the squared Mahalanobis distance

$$d_{ij}^2 = \tilde{\mathbf{r}}_{ij}^\top \tilde{\mathbf{P}}_{ij} \tilde{\mathbf{r}}_{ij}, \quad (4)$$

which replaces dot-product similarity with a consistency test under the precision prior induced by the SDE.

Robust estimation. The estimator in Eq. 3 assumes residuals are well-described by the predicted covariance. To improve robustness under model mismatch, we instead use a heavy-tailed likelihood over the Mahalanobis distance d_{ij}^2 , yielding a robust penalty $\rho(d_{ij}^2)$, as in M-estimation.

Minimizing this loss yields data-dependent influence weights:

$$w_{ij} \propto \frac{\partial}{\partial d_{ij}^2} \rho(d_{ij}^2),$$

which re-weight the precisions:

$$\mathbf{P}_{ij} \rightarrow w_{ij} \mathbf{P}_{ij},$$

reducing the influence of tokens whose residuals are unexpectedly large under the predicted covariance.

Two standard choices are:

$$w_{ij} \propto \begin{cases} \exp\left(-\frac{d_{ij}^2}{d}\right) & (\text{exponential}) \\ \left(1 + \frac{d_{ij}^2}{\nu}\right)^{-\kappa} & (\text{power law}) \end{cases} \quad (5)$$

where ν governs the tail weight of the influence function, $\kappa = \frac{\nu+d}{d}$, and d is the feature dimension. The exponential form recovers the standard dot-product attention structure, while the power-law form yields a heavier-tailed robust variant. As shown in Appendix A.4, these kernels correspond to Gaussian and Student- t residual models, respectively. Here, we use the power-law form since it defines a wider class, recovering the exponential form in the limit $\nu \rightarrow \infty$.

Closed-form parallel computation. To evaluate $\mathbf{P}_{ij}$ in closed form for all token pairs simultaneously, we require that the system matrices be simultaneously diagonalizable by some $\mathbf{S} \in \mathbb{C}^{d \times d}$: $\mathbf{A} = \mathbf{S} \mathbf{\Lambda} \mathbf{S}^{-1}$, $\mathbf{Q} = \mathbf{S} \mathbf{\Lambda}_Q \mathbf{S}^\dagger$, $\mathbf{R}_c = \mathbf{S} \mathbf{\Lambda}_R \mathbf{S}^\dagger$. Under this assumption, the DLE decouples into independent scalar ODEs for each eigenmode $\Sigma_{ij} = \mathbf{S} \mathbf{\Lambda}_{\Sigma,ij} \mathbf{S}^\dagger$, yielding a diagonal covariance,

$$\mathbf{\Lambda}_{\Sigma,ij} = \mathbf{\Lambda}_Q \frac{1 - e^{2 \operatorname{Re}(\mathbf{\Lambda}) \Delta t_{ij}}}{-2 \operatorname{Re}(\mathbf{\Lambda})} + \mathbf{\Lambda}_R e^{2 \operatorname{Re}(\mathbf{\Lambda}) \Delta t_{ij}} + \sigma_0^2 \mathbf{I}.$$

Likewise, $\mathbf{\Lambda}_{\tilde{\Sigma},ij} = \mathbf{\Lambda}_{\Sigma,ij} + \mathbf{\Lambda}_R$, and the precisions are obtained by diagonal inversion, $\mathbf{\Lambda}_{P,ij} = \mathbf{\Lambda}_{\tilde{\Sigma},ij}^{-1}$, $\mathbf{\Lambda}_{\tilde{P},ij} = \mathbf{\Lambda}_{\tilde{\Sigma},ij}^{-1}$. The Mahalanobis distance then decomposes into independent scalar contributions per mode in the eigenbasis. Letting $\hat{\mathbf{x}}_{s,j} = \mathbf{S}^{-1} \mathbf{C}^{-1} \mathbf{z}_j$, the robust precision-weighted aggregation takes the form:

$$\bar{\mathbf{x}}_{s,i} = \sum_{j \leq i} \mathcal{A}_{ij} \odot \hat{\mathbf{x}}_{s,ij}, \quad \hat{\mathbf{x}}_{s,ij} = e^{\mathbf{\Lambda} \Delta t_{ij}} \hat{\mathbf{x}}_{s,j},$$

$$\mathcal{A}_{ij} := w_{ij} \mathbf{\Lambda}_{P,ij} \left(\sum_{j' \leq i} w_{ij'} \mathbf{\Lambda}_{P,ij'} \right)^{-1},$$

Hence, the estimate can be computed cheaply in the eigenbasis. We can then recover an estimated measurement via $\bar{\mathbf{z}}_i = \mathbf{C} \mathbf{S} \bar{\mathbf{x}}_{s,i}$. When the eigenvalues of $\mathbf{A}$ are purely imaginary, $e^{\mathbf{\Lambda} \Delta t}$ reduces to element-wise rotations, as in RoPE.

3.2. Robust Filter Attention Mechanism

We instantiate the robust state estimator as a complex-valued attention layer by identifying the abstract diagonalization matrices with learned linear projections. The input projections $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v \in \mathbb{C}^{d \times d}$ learn the transformation into the diagonalizing basis of the SDE's system matrices, absorbing the inverse diagonalizing matrix $\mathbf{S}^{-1} \mathbf{C}^{-1}$, while

the output matrix W_o absorbs CS , mapping the filtered estimates back to the original basis:

$$Q = W_q Z, \quad K = W_k Z, \quad V = W_v Z \in \mathbb{C}^{d \times N}.$$

Separate query/key projections can be justified as arising from separate observation models (Appendix C.1).

To preserve the $\mathcal{O}(N^2 + Nd)$ memory complexity of standard attention, we impose isotropic decay and noise in the learned eigenbasis (per head):

$$\Lambda = -\mu I + i\Lambda_\Omega, \quad \Lambda_Q = \sigma^2 I, \quad \Lambda_R = \eta^2 I,$$

where $\mu, \sigma^2, \eta^2 \in \mathbb{R}^+$ and $\Lambda_\Omega \in \mathbb{R}^{d \times d}$ is diagonal with k th diagonal entry ω_k . This removes the ability to model dimension-dependent noise, but preserves the temporal dependence of uncertainty on lag. In practice, we introduce separate parameters for key-side (η^2), and query-side (γ^2) measurement noise (Appendix C.1).

Under isotropic decay and noise, each eigenmode follows independent exponentially decaying rotations with decay rate μ and angular frequency ω_k . This yields simple element-wise rotation factors for forward/backward propagation, and a decay kernel that depends only on the time lag Δt_{ij} :

$$\tilde{\Phi}^-[k, i] := e^{-i\omega_k t_i}, \quad \tilde{\Phi}^+[k, i] := e^{i\omega_k t_i}, \quad E[i, j] := e^{-\mu \Delta t_{ij}}.$$

We define rotated queries, keys, and values:

$$\tilde{Q} := \tilde{\Phi}^- \odot Q, \quad \tilde{K} := \tilde{\Phi}^- \odot K, \quad \tilde{V} := \tilde{\Phi}^- \odot V.$$

The isotropic constraints cause the variance to become independent of the feature dimension:

$$\begin{aligned} \Sigma_{\Delta t}[i, j] &:= \tilde{\sigma}^2(1 - E^2[i, j]) + \eta^2 E^2[i, j] + \sigma_0^2, \\ \tilde{\Sigma}_{\Delta t}[i, j] &:= \Sigma_{\Delta t}[i, j] + \gamma^2. \end{aligned} \quad (6)$$

Here, $\tilde{\sigma}^2 := \frac{\sigma^2}{2\mu}$, η^2 , γ^2 , σ_0^2 are learned scalar parameters (per head), corresponding, respectively, to steady-state process uncertainty, key/query measurement noise, and a noise floor. Note that $\Sigma_{\Delta t}$ denotes the scalar variance as a function of lag rather than a covariance matrix. Collecting terms,

$$\Sigma_{\Delta t}[i, j] = \alpha e^{-2\mu \Delta t_{ij}} + \beta, \quad \alpha := \eta^2 - \tilde{\sigma}^2, \quad \beta := \sigma_0^2 + \tilde{\sigma}^2.$$

The behavior of the variance with temporal lag is governed by the sign of α . When $\alpha < 0$, process noise dominates and precision decreases monotonically, yielding a *diffusive regime*. When $\alpha > 0$, measurement noise dominates and precision increases with lag, corresponding to an *integrative regime*. Because α and β are determined by learned scalar parameters per head, different heads can specialize into distinct filtering behaviors.

Letting $P_{\Delta t}[i, j] := \Sigma_{\Delta t}^{-1}[i, j]$ and $\tilde{P}_{\Delta t}[i, j] := \tilde{\Sigma}_{\Delta t}^{-1}[i, j]$, the isotropic constraint allows the Mahalanobis distance for

all pairs (i, j) to be computed by element-wise multiplying a matrix of scalar precisions by a matrix of squared residual norms $\|R_{qk}[i, j]\|^2$:

$$D^2[i, j] = \tilde{P}_{\Delta t}[i, j] \cdot \|R_{qk}[i, j]\|^2,$$

where the ij th residual is:

$$R_{qk}[i, j] := \tilde{Q}_i - E[i, j] \cdot \tilde{K}_j.$$

The squared residual norm decomposes into a query magnitude term, a decayed key magnitude term, and a cross-term containing the complex inner product:

$$\begin{aligned} \|R_{qk}[i, j]\|^2 &= \|Q_i\|^2 + E[i, j]^2 \cdot \|K_j\|^2 \\ &\quad - 2 E[i, j] \cdot \text{Re}(\tilde{Q}_i^\dagger \tilde{K}_j). \end{aligned} \quad (7)$$

Setting $\kappa = \frac{\nu+d}{d}$, $\nu = \nu_s d$, with $\nu_s \in \mathbb{R}^+$ yields dimension-normalized attention logits. Choosing the exponential form for w_{ij} in Eq. 5 yields logits

$$L = \log(P_{\Delta t}) - \frac{1}{\nu_s d} \tilde{P}_{\Delta t} \odot \|R_{qk}\|^2,$$

while the power-law form yields

$$L = \log(P_{\Delta t}) - (\nu_s + 1) \log\left(1 + \frac{1}{\nu_s d} \tilde{P}_{\Delta t} \odot \|R_{qk}\|^2\right).$$

The attention matrix is then $\hat{A} = A \odot E$, where:

$$A = \text{Softmax}_j(\beta_s L + M_{\text{causal}}),$$

where β_s is an additional inverse temperature parameter and $M_{\text{causal}} \in \{0, -\infty\}^{N \times N}$ is a causal mask.

The value estimate is computed by aggregating the rotated values and rotating the result back into the original value frame:

$$\bar{V} = \tilde{\Phi}^+ \odot (\tilde{V} \hat{A}^\top).$$

This rotate-aggregate-counter-rotate structure is required by the dynamical model: values from different time steps must be brought into a common temporal frame before aggregation, and the counter-rotation restores the output to the original frame.

The attention layer then computes an innovation step in the value basis,

$$\Delta V = \bar{V} - V,$$

which represents a correction from the current value toward the filtered estimate. This correction is projected back into the original basis and added to the residual stream:

$$Z^+ = Z + W_o \Delta V.$$

3.3. Real-valued Implementation

Although RFA is formulated over $\mathbb{C}^d$, all operations reduce to standard real arithmetic, as detailed in Appendix D.1. The complex projections $\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v \in \mathbb{C}^{d \times d}$ are implemented as real $d \times 2d$ matrices and $\mathbf{W}_o$ as a $2d \times d$ matrix. The complex rotations $\tilde{\Phi}^-$ and $\tilde{\Phi}^+$ correspond exactly to RoPE rotation and counter-rotation. RoPE corresponds to zero decay ($\mu = 0$), so that the complex exponential $e^{\Lambda \Delta t}$ reduces to pure rotation and the decay kernel $\mathbf{E}$ drops out.

Under the isotropic constraint, RFA preserves the asymptotic complexity of standard attention. The dominant operation remains a single $\mathcal{O}(N^2 d)$ matrix multiplication to compute the cross-term $\text{Re}(\tilde{\mathbf{Q}}^\dagger \tilde{\mathbf{K}})$. The remaining components—the decay kernel $\mathbf{E}[i, j]$ and precision kernel $\mathbf{P}_{\Delta t}[i, j]$ —are computed via elementwise operations with $\mathcal{O}(N^2)$ cost, and do not change the asymptotic complexity.

The complete implementation of the Isotropic RFA mechanism is formalized in Algorithm 1 in Appendix D.3.¹

3.4. Layer Depth as Iterative State Estimation

The robust M-estimator defines a fixed-point problem: the weights w_{ij} depend on residuals computed relative to the unknown latent state, which must be approximated by the current estimate. A single attention layer performs one such reweighting step using the previous layer’s output as the current state estimate, while residual connections implement a partial step toward each reweighted estimate.

Stacking layers therefore yields an iteratively reweighted least squares (IRLS)-like procedure, in which observations are progressively reweighted according to their consistency with the evolving estimate. This provides an interpretation of both depth and residual connections in RFA as successive refinement of the state estimate (Appendix C.6).

3.5. Filtering Behaviors

The uncertainty model influences the attention weights through two coupled mechanisms. The additive bias $\mathbf{B}_{\Delta t} := \log(\mathbf{P}_{\Delta t})$ acts as a prior budget allocated to tokens at each lag, while the multiplicative gate $\tilde{\mathbf{P}}_{\Delta t}$ controls the selectivity of attention — how sharply the model discriminates between consistent and inconsistent tokens at that lag. Both are determined by the learned noise parameters $(\tilde{\sigma}^2, \eta^2, \gamma^2, \sigma_0^2)$ through the scalar variance $\Sigma_{\Delta t} = \alpha e^{-2\mu \Delta t} + \beta$.

The sign of $\alpha = \eta^2 - \tilde{\sigma}^2$ defines a phase transition between two qualitatively distinct filtering behaviors:

Diffusive Regime ($\alpha < 0$). When process noise dominates

¹A multi-head implementation is available at <https://github.com/PCR-git/Robust-Filter-Attention>.

($\tilde{\sigma}^2 > \eta^2$), uncertainty accumulates monotonically with lag. The precision $\mathbf{P}_{\Delta t}$ acts as a *closing gate*: selectivity is maximal near the diagonal and degrades as lag increases. The additive bias decays toward a floor of $-\log(\beta)$, implementing a forgetting prior that suppresses distant tokens. These heads implement a recency bias, analogous to the linear distance penalties used in ALiBi (Appendix B.4).

Integrative Regime ($\alpha > 0$). When measurement noise dominates ($\eta^2 > \tilde{\sigma}^2$), the precision acts as an *opening gate*: selectivity is low near the diagonal and increases with lag as the initial measurement noise on the transported key dissipates under the stable dynamics. The additive bias correspondingly starts low and curves upward, implementing a settling prior that delays commitment until the transported observation becomes reliable. These heads function as lag-selective denoising filters, suppressing transient noise to identify stable historical structure.

Because α is determined by learned scalars per head, different heads can self-organize into different regimes during training. Explicit functional forms for the bias and gate in each regime are derived in Appendix B.4.

3.6. Recovery of Standard Positional Encodings

If the queries and keys are normalized, the Mahalanobis distance reduces to a dot product between queries and keys. In the zero-decay ($\mu = 0$), short-lag limit ($\Delta t \ll \beta/\sigma^2$), the RFA additive bias reduces to:

$$\mathbf{B}_{\Delta t} \approx -\frac{\sigma^2}{\beta} \Delta t + \text{const},$$

recovering a linear distance penalty consistent with the form of ALiBi’s linear bias (Appendix B.3.5).

Standard dot production attention with rotary positional encoding corresponds to the special case in which the dynamics are noiseless ($\sigma^2 = 0$), decay is absent ($\mu = 0$), query and key are normalized, and value rotation is omitted. Under these conditions, the state transition matrix reduces to a complex rotation and the precision prior is uniform.

3.7. Spectrally Coupled RFA (SC-RFA)

Standard positional mechanisms such as RoPE utilize a fixed frequency bank across all heads, allowing high-frequency oscillations to persist indefinitely. At long horizons, this leads to phase wrap-around: tokens separated by a full oscillation period produce similar phase configurations, making them difficult to distinguish based on position alone.

To address this, we introduce Spectrally Coupled RFA (SC-RFA), in which decay rates are coupled to frequencies across heads. We partition a global frequency bank Ω monotonically across heads, assigning each head h a spectral band $[\omega_{h,\min}, \omega_{h,\max}]$, and couple each head’s decay rate to its

maximum frequency:

$$\mu_h = b \cdot \omega_{h,\max},$$

where $b \in \mathbb{R}^+$ is a dimensionless damping coefficient (Appendix B.5). This enforces a fixed decay per oscillation cycle: over one period, the signal is attenuated by a factor of e^{-b} , directly controlling the trade-off between spectral resolution and long-range stability.

Each attention head induces a prior temporal response given by the product of a decay term and a precision term:

$$P_{\Delta t} \cdot e^{-\mu \Delta t} \propto \frac{e^{-\mu \Delta t}}{\alpha e^{-2\mu \Delta t} + \beta},$$

where $\alpha = \eta^2 - \tilde{\sigma}^2$ and $\beta = \sigma_0^2 + \tilde{\sigma}^2$.

In the integrative regime ($\alpha > 0$), precision initially increases with lag as key-side measurement noise dissipates, competing with exponential decay. The product peaks at a characteristic lag:

$$\Delta t^* = \frac{1}{2\mu} \log\left(\frac{\alpha}{\beta}\right).$$

Since μ_h varies across heads, peak locations $\Delta t_h^* \propto 1/\mu_h$ span a range of temporal scales, forming a bank of lag-selective filters. For sufficiently small μ , the peak may lie beyond the training context length, causing the integrative profile to appear as a monotonically increasing function over the training window and inducing an anti-recency bias that extrapolates poorly. Diffusive heads ($\alpha < 0$) provide a fallback by enforcing monotonic decay and stabilizing long-range behavior.

4. Experimental Evaluation and Ablations

We evaluate whether explicitly modeling uncertainty growth improves long-context stability while preserving short-range accuracy, comparing RFA against two widely used positional baselines derived from deterministic geometry (RoPE) (Su et al., 2024) and monotonic recency biasing (ALiBi) (Press et al., 2022), respectively. The per-head scalar parameters ($\tilde{\sigma}^2, \eta^2, \gamma^2, \sigma_0^2, \nu_s, \beta_s$) are learned entirely within the training window ($L = 512$), and evaluated under zero-shot extrapolation at longer context lengths ($L \in \{512, 1024, 2048, 4096\}$).

4.1. Experimental Setup

Architecture. All models use a 6-layer Transformer with $h = 8$ heads and embedding dimension $d = 256$. To ensure comparable model capacity, we apply identical $d \rightarrow 2d \rightarrow d$ projections in both RFA and the RoPE/ALiBi baselines, which represent the mapping between the real and complex domains (Appendix D.1). RFA introduces only

a small number of additional scalar parameters per head for noise and robustness, increasing total parameter count by approximately 0.02%. We employ a pre-norm architecture with an FFN expansion factor of 4. Models are trained for 15 epochs until convergence using Adam with a cosine learning rate schedule.

Datasets. We evaluate on WikiText-103, a large-scale word-level language modeling benchmark derived from Wikipedia articles and used to measure perplexity and long-context extrapolation (Merity et al., 2017), and on BabyLM-2025 (Strict), a curated English language modeling corpus used as a complementary benchmark under the same training and evaluation protocol (Charpentier et al., 2025).

Ablations. We compare against standard positional baselines: RoPE (B1) and ALiBi (B2), and include two geometry-only decay variants to isolate the effect of damping in rotational embeddings: Decayed RoPE (B3), which applies exponential decay with distance, as in RFA, and SC-RoPE (B4), which couples decay rates to head-wise frequency bands, as in SC-RFA. These baselines test whether decay and spectral coupling alone can explain extrapolation gains, without modeling uncertainty.

We evaluate RFA (M1) and two variants of SC-RFA: one optimized for in-window performance (M2), with $b = 0.05$, and one optimized for long-context extrapolation via increased damping (M3), with $b = 5.0$. We also include structural ablations relative to M2, designed to isolate the effect of its components when removed: the power-law robust weight w_{ij} , replacing it with an exponential weight (M2.1); the DLE-derived precision prior (M2.2); the multiplicative gating term $P_{\Delta t}$ (M2.3); value rotations (M2.4); all rotations (M2.5); finally, we test a purely rotational, zero decay and zero noise variant, analogous to RoPER (M2.6).

Full architectural and ablation details are provided in Appendix E. Analysis of attention maps and noise parameters are provided in Appendix F.

4.2. Results on Wikitext-103

We evaluate extrapolation by measuring test perplexity on WikiText-103 at increasing context lengths, after training all models with a fixed context window of 512 tokens. Results are shown in Table 1.

RFA variants achieve both stronger local performance and improved extrapolation relative to RoPE. In particular, SC-RFA (M2) improves over RoPE by 0.94 PPL at $L = 512$ and reduces degradation at long horizons, reaching 37.19 PPL at $L = 4096$ compared to RoPE’s 72.69. M2 (SC-RFA) outperforms M1 (RFA) across all context lengths.

Unlike RoPE, which degrades monotonically outside the training window, RFA exhibits a non-monotonic extrap-

Table 1. Long-context extrapolation on WikiText-103 (Test PPL). All models were trained with a fixed context window of 512 tokens.

Model	L=512	L=1024	L=2048	L=4096
RoPE (B1)	28.48	30.94	44.21	72.69
ALiBi (B2)	28.59	27.30	26.54	26.30
Decayed RoPE (B3)	28.45	30.45	40.03	64.54
SC-RoPE (B4)	28.44	30.49	41.06	60.08
RFA (M1)	28.01	27.58	29.99	38.46
SC-RFA (M2)	27.54	26.73	29.46	37.19
SC-RFA (M3)	27.91	26.68	26.37	28.16
<i>Structural Ablations (Relative to M2)</i>				
Exp. Weight (M2.1)	27.98	27.16	28.95	33.51
Flat Prior (M2.2)	27.69	28.71	38.11	62.83
No Mult. Gate (M2.3)	27.65	29.01	39.18	57.30
No Value Rot. (M2.4)	30.24	92.08	187.29	463.29
No Rotations (M2.5)	28.58	27.25	26.61	26.83
Pure Rotation (M2.6)	27.97	35.59	69.39	131.29

olation profile: perplexity initially decreases beyond the training horizon before increasing at longer context lengths.

With higher damping coefficient, SC-RFA (M3) further improves long-context stability, achieving nearly flat perplexity at up to $L = 4096$ while maintaining competitive performance within the training window. This behavior emerges under a fixed training protocol without requiring length-dependent scaling rules or curriculum schedules.

Introducing decay into rotational embeddings (B3) and spectrally coupling decay across heads (B4) slows the long-range degradation of RoPE. However, both geometry-only variants underperform RFA across all context lengths, indicating that decay alone is insufficient without explicit uncertainty modeling. B4 does not improve substantially over B3, indicating that coupling decay to the frequency spectrum alone provides little additional benefit.

The exponential weight variant of SC-RFA (M2.1) underperforms the power-law robust weighting at the training horizon, but achieves lower perplexity at extreme extrapolation lengths. This is consistent with Gaussian likelihoods imposing stronger quadratic penalties on residuals, which suppress extreme deviations more aggressively but reduce sensitivity to small errors when uncertainty is low.

Removing the DLE-derived precision prior (M2.2) leads to degradation at long horizons, with perplexity increasing to 62.83 at $L = 4096$, demonstrating that explicit uncertainty propagation via the DLE is necessary for stable long-context behavior. Removing the precision term $P_{\Delta t}$ from the Mahalanobis distance (M2.3) causes degradation within the training window and worsens extrapolation, indicating that both the additive and multiplicative precision terms contribute to stability.

Table 2. Sensitivity analysis of the damping coefficient b in SC-RFA (M2), with RoPE (B1) and ALiBi (B2) as baselines. Results show Test PPL on WikiText-103 across increasing context lengths.

Damping (b)	L=512	L=1024	L=2048	L=4096
RoPE (B1)	28.48	30.94	44.21	72.69
ALiBi (B2)	28.59	27.30	26.54	26.30
5×10^{-4}	27.60	28.88	37.34	51.48
5×10^{-3}	27.60	28.71	35.35	43.90
5×10^{-2}	27.54	26.73	29.46	37.19
5×10^{-1}	27.61	26.38	26.37	29.72
5×10^0	27.91	26.68	26.37	28.16

Eliminating value-space rotation and counter-rotation (M2.4) causes severe degradation at long context, reaching 463.29 PPL at $L = 4096$. This is consistent with aggregation no longer corresponding to fusion of latent state estimates in a shared temporal frame. Removing all rotations (M2.5) degrades short-context performance but yields strong long-range stability. In this setting, RFA reduces to a purely distance-dependent bias with logarithmic scaling, closely matching the behavior and performance of ALiBi.

In the zero-noise, zero-decay, pure rotational setting (M2.6), perplexity increases sharply with context length. Despite outperforming RoPE within the training window, value rotations without decay and uncertainty modeling accelerate long-range degradation rather than preventing it.

Compared to ALiBi, SC-RFA achieves lower perplexity at the training length ($L = 512$) and at moderate extrapolation ($L = 1024$), suggesting improved utilization of fine-grained temporal structure when uncertainty remains low. At longer horizons, ALiBi attains lower perplexity by enforcing strict locality, effectively suppressing long-range interactions, while SC-RFA continues to integrate distant context with attenuated but nonzero precision. This reflects a trade-off between aggressive locality and uncertainty-weighted long-range integration.

Table 2 shows that this tradeoff is continuously controlled in SC-RFA by the damping coefficient b . Smaller values of b yield slower decay, improving short-context performance but leading to faster degradation as context increases. Larger values of b produce stronger attenuation and more stable long-range behavior at the cost of reduced short-range performance. Notably, for sufficiently strong damping ($b \geq 0.5$), SC-RFA outperforms ALiBi at intermediate horizons ($L = 2048$), with ALiBi retaining an advantage only at the largest tested context length.

4.3. Results on BabyLM-2025

We use the same architectures, hyperparameters, and training protocol as on WikiText-103. On BabyLM-2025, where

Table 3. Long-context extrapolation on BabyLM-2025 (Test PPL). All models were trained with a fixed context window of 512 tokens.

Model	L=512	L=1024	L=2048	L=4096
RoPE (B1)	17.70	18.78	23.33	33.29
ALiBi (B2)	17.70	17.20	17.06	17.51
RFA (M1)	17.51	17.71	20.61	31.04
SC-RFA (M2)	17.36	16.99	18.33	22.25
SC-RFA (M3)	17.51	17.07	17.26	18.74

language modeling performance is more strongly dominated by short-range context, differences between positional mechanisms are smaller at short context lengths. The RFA variants outperform RoPE at all evaluated context lengths and outperform ALiBi within the training window. SC-RFA also achieves lower perplexity than ALiBi at intermediate context ($L = 1024$), while ALiBi remains strongest at the longest horizons due to its strict recency bias. These results mirror the trade-off observed on WikiText-103: precision weighting improves robustness over purely rotational embeddings while retaining stronger short- and mid-range performance than aggressively local recency biases.

4.4. Learning Dynamics and Head Specialization

RFA variants achieve lower validation perplexity earlier in training than RoPE and ALiBi, indicating that the SDE-based prior provides an effective inductive bias for latent state estimation (Appendix F.1). Analysis of learned noise, decay, and robustness parameters reveals systematic specialization across heads into distinct uncertainty and selectivity regimes (Appendix F.2).

Attention map visualizations at long context lengths reveal structured multi-scale behavior in RFA. Compared to RoPE, attention maps exhibit reduced checkerboard interference from high-frequency oscillations and more coherent periodic structure, reflecting aggregation in a dynamically consistent frame. Some heads exhibit an integrative regime, in which attention is initially suppressed near the diagonal and peaks at a characteristic lag, indicating delayed aggregation until the latent state estimate stabilizes (Appendix F.3).

SC-RFA sharpens and organizes this structure through spectral coupling, inducing a clear ordering of temporal specialization across heads. High-decay heads concentrate attention near the diagonal, while lower-decay heads shift their mass toward progressively longer temporal offsets, producing distinct lag bands. This results in fewer and sharper periodic bands and a clearer separation between local and long-range interactions. Together, these patterns support the interpretation of RFA as learning a structured, uncertainty-aware temporal filter bank rather than relying solely on geometric positional bias.

5. Conclusion

We reformulate self-attention as a tractable approximation to state estimation under a linear stochastic dynamical model, yielding Robust Filter Attention (RFA). In this formulation, attention weights reflect uncertainty-aware agreement between dynamically transported representations rather than static feature similarity, while preserving the computational structure of standard attention. RFA recovers existing positional mechanisms as limiting cases and improves temporal consistency and long-context behavior.

Tractability requires four structural constraints — a factorized likelihood, linear time-invariant dynamics, simultaneous diagonalizability, and isotropic noise and decay per head. A central challenge for future work is relaxing these constraints while retaining tractable covariance propagation and parallel computation.

Other important directions include understanding how these dynamical and uncertainty-based mechanisms interact with architectural components such as normalization and depth, as well as how similar uncertainty-aware formulations can be incorporated into other sequence modeling architectures.

Impact Statement

This work introduces an uncertainty-aware formulation of self-attention. We do not identify any ethical concerns beyond those generally associated with advances in machine learning methodology.

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A. Attention from a Batch State Estimation Formulation

Here, we derive the RFA mechanism introduced in Section 3 in greater detail. We briefly recall the generative model and notation. The latent state evolves under the linear SDE

$$dx(t) = Ax(t)dt + Gdw(t), \quad z_i = Cx(t_i) + v_i,$$

where $v_i \sim \mathcal{N}(0, R)$ and $Q = GG^\top$. Each past observation is mapped into the latent space and propagated to the query position:

$$\hat{x}_{ij} = e^{A\Delta t_{ij}} C^{-1} z_j, \quad \Delta t_{ij} = t_i - t_j.$$

Under the SDE,

$$\hat{x}_{ij} = x(t_i) + r_{ij}, \quad r_{ij} \sim \mathcal{N}(0, \Sigma_{ij}),$$

with covariance

$$\Sigma_{ij} = V(\Delta t_{ij}) + e^{A\Delta t_{ij}} R_c e^{A^\top \Delta t_{ij}},$$

where $R_c = C^{-1}RC^{-\top}$, and $V(\Delta t)$ is the DLE solution derived below. Figure 1 illustrates this propagation process for a stable two-dimensional LTI system.

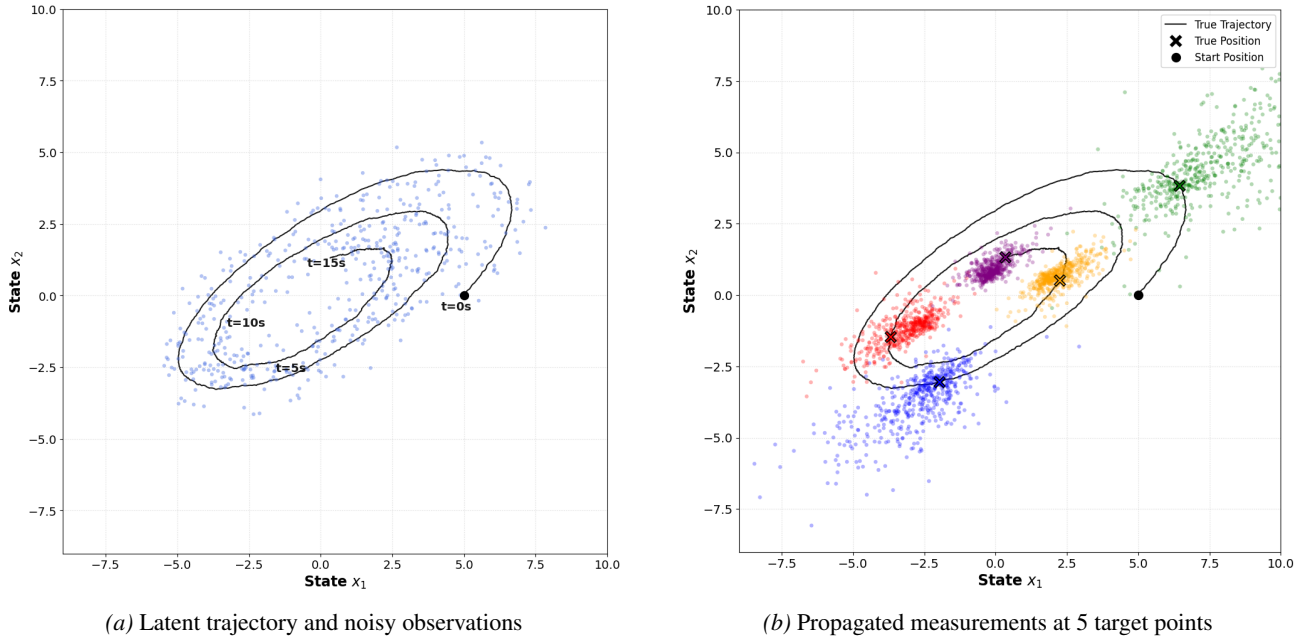


Figure 1. Illustration of an LTI SDE in 2D. (a) The true trajectory (black) is observed through additive Gaussian measurement noise $v_t \sim \mathcal{N}(0, R)$; noisy measurements z_i are shown as blue points. (b) For five target times t_i , the plot shows the ensemble of transported keys $\hat{x}_{ij}$ mapped through the transition $e^{A\Delta t_{ij}}$ from all noisy measurements z_j (where we have used $C = I$). Both forward and backward transported $\hat{x}_{ij}$ are shown for visualization; estimation is causal and uses only $j \leq i$.

A.1. Analytical Solution of the Differential Lyapunov Equation (DLE)

For parallel aggregation across all token pairs, we must efficiently solve the DLE for all $i, j \in [1, N]$. To obtain an analytically tractable solution, we assume the system matrices are simultaneously diagonalizable by an invertible $S \in \mathbb{C}^{d \times d}$, where $A = SAS^{-1}$ and $Q = SA_Q S^\dagger$. This assumption corresponds to learning dynamics in a basis of decoupled modes.

The propagated state covariance, $V(\Delta t_{ij})$, is the solution to the DLE (Eq. 2):

$$V(\Delta t_{ij}) = \int_0^{\Delta t_{ij}} e^{As} Q e^{A^\top s} ds.$$

Transforming to the eigenbasis, the covariance becomes:

$$V(\Delta t_{ij}) = SA_V(\Delta t_{ij})S^\dagger.$$

where each diagonal entry of $\Lambda_V(\Delta t_{ij})$ satisfies the scalar integral:

$$\lambda_{V,k}(\Delta t_{ij}) = \lambda_{Q,k} \int_0^{\Delta t_{ij}} e^{(\lambda_k + \lambda_k^*)s} ds = \lambda_{Q,k} \int_0^{\Delta t_{ij}} e^{2\text{Re}(\lambda_k)s} ds$$

(where $\Lambda_V = \text{diag}(\lambda_V)$ and $\Lambda_Q = \text{diag}(\lambda_Q)$). Each mode accumulates noise according to its real decay rate $\text{Re}(\lambda_k)$.

Evaluating this integral yields the analytical solution $\varphi(\lambda, \lambda_Q, \Delta t)$ (for the causal case $\Delta t \geq 0$):

$$\Lambda_V(\Delta t_{ij}) = \text{diag}(\varphi(\lambda_k, \lambda_{Q,k}, \Delta t_{ij}))_{k=1}^d,$$

$$\varphi(\lambda, \lambda_Q, \Delta t) = \begin{cases} \lambda_Q \frac{1 - e^{2\text{Re}(\lambda) \Delta t}}{-2\text{Re}(\lambda)}, & \text{Re}(\lambda) \neq 0, \\ \lambda_Q \Delta t, & \text{Re}(\lambda) = 0. \end{cases}$$

Assuming that $R_c = S\Lambda_R S^\dagger$, the covariance becomes:

$$\Sigma_{ij} = S\Lambda_{\Sigma,ij}S^\dagger, \quad \Lambda_{\Sigma,ij} = \Lambda_V(\Delta t_{ij}) + \Lambda_R e^{2\text{Re}(\Lambda)\Delta t_{ij}} + \sigma_0^2 I,$$

where we have added a small noise floor σ_0^2 . This covariance is bounded for all $\Delta t_{ij} \geq 0$ if and only if $\text{Re}(\lambda_k) < 0$ for all k , i.e., the dynamics are stable. The residual $r_{ij} = \hat{x}_i - \hat{x}_{ij}$ has covariance $\tilde{\Sigma}_{ij} = S\Lambda_{\tilde{\Sigma},ij}S^\dagger$, where $\Lambda_{\tilde{\Sigma},ij} = \Lambda_{\Sigma,ij} + \Lambda_R$. The precision is then obtained by diagonal inversion:

$$P_{ij} = S^{-\dagger} \Lambda_{P,ij} S^{-1}, \quad \tilde{P}_{ij} = S^{-\dagger} \Lambda_{\tilde{P},ij} S^{-1},$$

$$\Lambda_{P,ij} = \Lambda_{\Sigma,ij}^{-1}, \quad \Lambda_{\tilde{P},ij} = \Lambda_{\tilde{\Sigma},ij}^{-1}.$$

The observable precision $\tilde{P}_{ij}$ is used to evaluate prediction consistency, whereas the latent precision P_{ij} determines how much each transported estimate contributes to the estimate.

A.2. Precision-Weighted Consensus State Estimation

To estimate the latent state $x(t_i)$, we consider the transported observations $\{\hat{x}_{ij}\}_{j \leq i}$, which are jointly Gaussian under the SDE. Because process noise accumulates along shared portions of the trajectory, these observations are generally correlated. The optimal estimator therefore requires inference under the full joint covariance, which entails inversion of a dense covariance matrix or recursive filtering.

To obtain a parallel attention-style estimator, we neglect the cross-covariances induced by shared process noise while preserving the marginal covariance of each transported observation. Under this approximation, each transported observation is treated as an independent noisy measurement of the latent state with its SDE-derived marginal covariance.

Let $x \in \mathbb{R}^d$ denote a candidate value of the latent state at time t_i . We model

$$p(\hat{x}_{ij} | x) = \mathcal{N}(\hat{x}_{ij}; x, \Sigma_{ij}),$$

where Σ_{ij} is the SDE-derived marginal covariance of the transported observation and $P_{ij} = \Sigma_{ij}^{-1}$ is the corresponding precision. Neglecting cross-covariances yields the factorized likelihood:

$$p(\{\hat{x}_{ij}\}_{j \leq i} | x) \approx \prod_{j \leq i} p(\hat{x}_{ij} | x).$$

Maximizing this likelihood is equivalent to the precision-weighted least-squares problem:

$$\bar{x}_i = \arg \min_x \sum_{j \leq i} (x - \hat{x}_{ij})^\top P_{ij} (x - \hat{x}_{ij}),$$

which yields the closed-form estimator:

$$\bar{x}_i = \left(\sum_{j \leq i} P_{ij} \right)^{-1} \sum_{j \leq i} P_{ij} \hat{x}_{ij}.$$

This approximation is exact when $\mathbf{Q} = \mathbf{0}$. In general, shared process noise induces cross-covariances between transported observations that the factorized likelihood ignores, causing correlated evidence to be overcounted. However, the marginal covariance of each transported observation is preserved, so the resulting precision weights remain individually consistent with the uncertainty predicted by the SDE. The approximation therefore sacrifices globally consistent inference in exchange for a tractable parallel attention-style estimator with structured uncertainty propagation.

The precision-weighted estimator above assumes a Gaussian likelihood over residuals. To improve robustness under model mismatch, we reweight each precision matrix by a scalar function of the squared Mahalanobis distance $w(d_{ij}^2)$,

$$\mathbf{P}_{ij} \rightarrow w_{ij} \mathbf{P}_{ij}, \quad d_{ij}^2 = \mathbf{r}_{ij}^\top \tilde{\mathbf{P}}_{ij} \mathbf{r}_{ij}.$$

This yields the reweighted estimator

$$\bar{\mathbf{x}}_i = \left(\sum_{j \leq i} w_{ij} \mathbf{P}_{ij} \right)^{-1} \sum_{j \leq i} w_{ij} \mathbf{P}_{ij} \hat{\mathbf{x}}_{ij}.$$

Typical choices include exponential weights, corresponding to a Gaussian likelihood, and power-law weights, corresponding to a Student- t likelihood:

$$w_{ij} \propto \begin{cases} \exp\left(-\frac{d_{ij}^2}{d}\right), \\ \left(1 + \frac{d_{ij}^2}{\nu}\right)^{-\kappa}. \end{cases}$$

Since the weights depend on the unknown state estimate through the residuals, the resulting estimator may be interpreted as a fixed-point problem and is closely related to iteratively reweighted least squares (IRLS), discussed further in Appendix C.6.

A.3. Parallel Aggregation via Diagonalization

To obtain a scalable implementation, we transform the robust precision-weighted average to the diagonalized basis. We define the state and propagated measurements in this basis as:

$$\hat{\mathbf{x}}_{s,i} := \mathbf{S}^{-1} \hat{\mathbf{x}}_i, \quad \hat{\mathbf{x}}_{s,ij} := e^{\Lambda \Delta t_{ij}} \hat{\mathbf{x}}_{s,j},$$

and the corresponding observable residual:

$$\tilde{\mathbf{r}}_{s,ij} = \hat{\mathbf{x}}_{s,i} - \hat{\mathbf{x}}_{s,ij}.$$

Using the simultaneous diagonalization, $\tilde{\mathbf{P}}_{ij} = \mathbf{S}^{-\dagger} \Lambda_{\tilde{\mathbf{P}},ij} \mathbf{S}^{-1}$, and the Mahalanobis distance decomposes into independent scalar components:

$$d_{ij}^2 = \tilde{\mathbf{r}}_{s,ij}^\dagger \Lambda_{\tilde{\mathbf{P}},ij} \tilde{\mathbf{r}}_{s,ij} = \sum_{k=1}^d \lambda_{\tilde{\mathbf{P}},ij,k} |\tilde{r}_{s,ij,k}|^2.$$

where $\lambda_{\tilde{\mathbf{P}},ij,k}$ and $\tilde{r}_{s,ij,k}$ are the k th diagonal components of $\Lambda_{\tilde{\mathbf{P}},ij}$ and $\tilde{\mathbf{r}}_{s,ij}$, respectively. This allows the robust weights $w_{ij} = w(d_{ij}^2)$ to be computed efficiently for all token pairs.

Applying the aggregation in this basis yields:

$$\bar{\mathbf{x}}_{s,i} = \left(\sum_{j \leq i} w_{ij} \Lambda_{\mathbf{P},ij} \right)^{-1} \sum_{j \leq i} w_{ij} \Lambda_{\mathbf{P},ij} \hat{\mathbf{x}}_{s,ij}.$$

Since all matrices are diagonal, both the sum and inverse are element-wise operations. We then define attention weights as:

$$\tilde{\mathcal{A}}_{ij} := w_{ij} \Lambda_{\mathbf{P},ij}, \quad \mathcal{A}_{ij} = \tilde{\mathcal{A}}_{ij} \odot \left(\sum_{j' \leq i} \tilde{\mathcal{A}}_{ij'} \right),$$

where $\odot$ denotes element-wise division. The aggregation then takes the form:

$$\bar{\mathbf{x}}_{s,i} = \sum_{j \leq i} \mathcal{A}_{ij} \odot \hat{\mathbf{x}}_{s,ij},$$

where, instead of scalar attention weights, each $\mathcal{A}_{ij}$ is a vector that element-wise multiplies $\hat{\mathbf{x}}_{s,ij}$. Finally, the output in the original coordinate system is recovered by $\bar{\mathbf{z}}_i = \mathbf{C}\mathbf{S}\bar{\mathbf{x}}_{s,i}$. All operations are $\mathcal{O}(d)$ per token pair, yielding an overall complexity of $\mathcal{O}(N^2d)$ with no matrix inversions.

Equivalently, the normalized attention weights may be written in Softmax form. Since

$$\mathcal{A}_{ij,k} \propto \lambda_{P,ij,k} w_{ij},$$

the corresponding dimension-wise attention logits are

$$\mathcal{L}_{ij,k} = \log \lambda_{P,ij,k} + \log w_{ij}.$$

For exponential reweighting $w_{ij} \propto \exp(-d_{ij}^2/d)$, the corresponding logit becomes:

$$\mathcal{A}_{ij,k} = \frac{\exp(\mathcal{L}_{ij,k})}{\sum_{j' \leq i} \exp(\mathcal{L}_{ij',k})}, \quad \mathcal{L}_{ij,k} := \log \lambda_{P,ij,k} - \frac{1}{d} \lambda_{\tilde{P},ij,k} |\tilde{\mathbf{r}}_{s,ij,k}|^2.$$

For the power law influence function $w_{ij} \propto (1 + d_{ij}^2/\nu)^{-\kappa}$, the logit becomes:

$$\mathcal{L}_{ij,k} := \log \lambda_{P,ij,k} - \kappa \log \left(1 + \frac{1}{\nu} \lambda_{\tilde{P},ij,k} |\tilde{\mathbf{r}}_{s,ij,k}|^2 \right),$$

Remark. The normalization term $\left(\sum_{j' \leq i} \tilde{\mathcal{A}}_{ij'} \right)^{-1}$ corresponds to the approximate posterior covariance in the diagonalizing basis under the CI assumption, providing a measure of the model's confidence in the aggregated state estimate. This quantity is a byproduct of the formulation that goes unused in standard attention, and may be useful beyond weighting — for instance, to gate updates or signal when the context provides insufficient evidence. We leave exploration of such uses to future work.

A.4. Likelihood-Induced Attention Kernels (Isotropic Case)

Under the isotropic covariance constraint $\Sigma_{ij} = \Sigma^2(\Delta t_{ij}) \mathbf{I}_d$, the squared Mahalanobis distance reduces to $d_{ij}^2 = \frac{\|\tilde{\mathbf{r}}_{ij}\|^2}{\Sigma^2(\Delta t_{ij})}$. Different choices of the robust weighting function induce different attention kernels.

Gaussian likelihood. For an isotropic Gaussian likelihood, the dimension-normalized log likelihood is

$$\mathcal{L}_{ij}^{\text{Gauss}} = -\log \Sigma^2(\Delta t_{ij}) - \frac{1}{d} d_{ij}^2,$$

up to additive constants absorbed by Softmax normalization. Exponentiating therefore gives $w_{ij} \propto \exp(-\frac{d_{ij}^2}{d})$, recovering the dimension-normalized exponential attention kernel.

Student- t likelihood. Assuming instead an isotropic multivariate Student- t distribution, $\tilde{\mathbf{r}}_{ij} \sim t_\nu(\mathbf{0}, \tilde{\Sigma}^2(\Delta t_{ij}) \mathbf{I}_d)$, the dimension-normalized negative log-likelihood becomes

$$\mathcal{L}_{ij}^t = -\log \Sigma^2(\Delta t_{ij}) - \frac{\nu + d}{d} \log \left(1 + \frac{d_{ij}^2}{\nu} \right),$$

up to additive constants. The corresponding attention kernel is therefore $w_{ij} \propto \left(1 + \frac{d_{ij}^2}{\nu} \right)^{-\kappa}$, $\kappa = \frac{\nu+d}{d}$, where κ sets the rate at which large residuals are downweighted.

In the large- ν limit,

$$\lim_{\nu \rightarrow \infty} \left(1 + \frac{d_{ij}^2}{\nu} \right)^{-(\nu+d)/d} = \exp \left(-\frac{d_{ij}^2}{d} \right),$$

so that

$$\mathcal{L}_{ij}^t \longrightarrow \mathcal{L}_{ij}^{\text{Gauss}} + \text{const.}$$

The Student- t formulation therefore provides a family of attention kernels, with the dimension-normalized exponential kernel recovered in the large- ν limit and progressively heavier-tailed kernels obtained as the degrees of freedom decrease.

B. Robust Filter Attention Mechanism

We now instantiate the state estimation formulation as an attention mechanism, proceeding from the full anisotropic tensor formulation implied by the derivation to the memory-efficient isotropic variant used in practice.

B.1. Anisotropic Tensor RFA

Under diagonalizable dynamics, the most general RFA formulation propagates and weights each feature dimension independently, yielding an $\mathcal{O}(N^2d)$ attention tensor. This is not memory-scalable but serves as the reference from which the isotropic variant is derived.

Learned Change-of-Basis Projections. The transformation to the decoupled eigenbasis is learned through complex-valued projections. We define:

$$\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v, \mathbf{W}_o \in \mathbb{C}^{d \times d},$$

where d is the embedding dimension. The input projections $\{\mathbf{W}_q, \mathbf{W}_k, \mathbf{W}_v\}$ parameterize the learned diagonalizing basis $\mathbf{S}^{-1}\mathbf{C}^{-1}$, mapping the input into the eigenbasis of $\mathbf{A}$ where the DLE is analytically solvable, while the output projection $\mathbf{W}_o$ parameterizes $\mathbf{CS}$, mapping the filtered state estimate back into the original embedding space.

Given an input sequence $\mathbf{Z} \in \mathbb{R}^{d \times N}$, we obtain query, key, and value representations as usual:

$$\mathbf{Q} = \mathbf{W}_q \mathbf{Z}, \quad \mathbf{K} = \mathbf{W}_k \mathbf{Z}, \quad \mathbf{V} = \mathbf{W}_v \mathbf{Z}.$$

Key and Value Propagation. In this reference model, every feature k possesses its own complex eigenvalue λ_k . We define the propagation tensors $\mathcal{E}$ and the resulting propagated keys and values:

$$\mathcal{E}[k, i, j] = e^{\lambda_k(t_i - t_j)}, \quad \hat{\mathcal{K}}[k, i, j] = \mathcal{E}[k, i, j] \cdot \mathbf{K}[k, j], \quad \hat{\mathcal{V}}[k, i, j] = \mathcal{E}[k, i, j] \cdot \mathbf{V}[k, j].$$

Residuals & Precision. We compute the residual tensor $\mathcal{R}_{qk}$:

$$\mathcal{R}_{qk}[k, i, j] = \mathbf{Q}[k, i] - \hat{\mathcal{K}}[k, i, j],$$

This is weighted by the analytic precision tensor $\tilde{\mathcal{P}}$, which is defined element-wise for each channel k using the DLE solution, where $\mu_k := -\text{Re}(\lambda_k)$:

$$\begin{aligned} \Sigma[k, i, j] &= \tilde{\sigma}_k^2 (1 - e^{-2\mu_k \Delta t_{ij}}) + \eta_k^2 e^{-2\mu_k \Delta t_{ij}} + \sigma_0^2, & \tilde{\Sigma}[k, i, j] &= \Sigma[k, i, j] + \gamma_k^2, \\ \mathcal{P}[k, i, j] &= \Sigma[k, i, j]^{-1}, & \tilde{\mathcal{P}}[k, i, j] &= \tilde{\Sigma}[k, i, j]^{-1}. \end{aligned}$$

Aggregation. Unlike standard attention, which applies a single scalar score per head, tensor RFA computes an attention tensor $\mathcal{A} \in \mathbb{R}^{d \times N \times N}$. The logit tensor is then:

$$\mathcal{L}[k, i, j] = \log(\mathcal{P}[k, i, j]) - \kappa \log \left(1 + \frac{1}{\nu} \sum_{k'} \tilde{\mathcal{P}}[k', i, j] \cdot |\mathcal{R}_{qk}[k', i, j]|^2 \right).$$

The estimate in the eigenbasis is computed via a row-wise Softmax over the logits, followed by a weighted sum:

$$\mathcal{A}[k, i, j] = \text{Softmax}_j(\mathcal{L}[k, i, j]), \quad \bar{\mathbf{V}}[k, i] = \sum_{j \leq i} \mathcal{A}[k, i, j] \cdot \hat{\mathcal{V}}[k, i, j].$$

The time complexity remains $\mathcal{O}(N^2d)$, but storing the propagated keys and values, residuals, and attention tensors requires $\mathcal{O}(N^2d)$ memory, limiting scalability. We therefore derive a memory-efficient implementation that avoids storing tensors.

B.2. Factorization and Complexity Reduction

We introduce the following factorizations to simplify the computation.

B.2.1. TOEPLITZ KERNEL FOR PRECISION

If the measurements occur at equal time intervals δt , the covariance tensor $\Sigma[k, i, j]$ depends only on the channel k and the time lag $\tau = |i - j|$. This induces a Toeplitz structure along the temporal dimensions for each channel.

Letting $\Delta t_{ij} = \tau \delta t$, we can thus pre-compute a 1D covariance kernel: $\mathcal{K}_\Sigma \in \mathbb{R}^{d \times N}$:

$$\mathcal{K}_\Sigma[k, \tau] = \tilde{\sigma}_k^2 (1 - e^{-2\mu_k \delta t \tau}) + \eta_k^2 e^{-2\mu_k \delta t \tau} + \sigma_0^2,$$

The full precision tensor is then simply the element-wise inverse of this kernel:

$$\mathcal{P}[k, i, j] = \mathcal{K}_P[k, |i - j|] := 1/\mathcal{K}_\Sigma[k, |i - j|].$$

B.2.2. TRANSPORT THROUGH A COMMON REFERENCE FRAME

For an LTI system, the state transition depends only on the time difference, $e^{\Lambda(t_i - t_j)}$. Using the fact that $e^{\Lambda(t_i - t_j)} = e^{\Lambda t_i} e^{-\Lambda t_j}$, the propagation can be separated into two stages: a backward transport that maps every measurement into a common stationary reference frame ($t = 0$), followed by a forward transport into the query time.

Accordingly, we define the forward and backward transport factors

$$\Phi^+[k, i] := e^{\lambda_k t_i}, \quad \Phi^-[k, j] := e^{-\lambda_k t_j},$$

so that the propagation tensor factorizes as

$$\mathcal{E}[k, i, j] = \Phi^+[k, i] \Phi^-[k, j].$$

Rather than constructing a distinct transport operator for every pair (i, j) , each key, query, and value is transported only once into the stationary frame:

$$\hat{Q}[k, j] := \Phi^-[k, j] Q[k, j], \quad \hat{K}[k, j] := \Phi^-[k, j] K[k, j], \quad \hat{V}[k, j] := \Phi^-[k, j] V[k, j].$$

The propagated keys and values are then recovered by applying the query-dependent forward transport,

$$\hat{\mathcal{K}}[k, i, j] = \Phi^+[k, i] \hat{K}[k, j], \quad \hat{\mathcal{V}}[k, i, j] = \Phi^+[k, i] \hat{V}[k, j].$$

Since $\Phi^+[k, i]$ is independent of the source index j , it can be pulled outside the attention aggregation:

$$\bar{V}[k, i] = \sum_{j \leq i} \mathcal{A}[k, i, j] \hat{V}[k, i, j] = \Phi^+[k, i] \sum_{j \leq i} \mathcal{A}[k, i, j] \hat{V}[k, j].$$

Thus, transport is performed by mapping all representations into a common stationary frame, performing attention there, and applying a single forward transport to recover the output in the query frame. Figure 2 illustrates this decomposition.

B.2.3. MEMORY EFFICIENCY AND NUMERICAL STABILITY

Recall that:

$$\mathcal{R}_{qk}[k, i, j] = Q[k, i] - \hat{\mathcal{K}}[k, i, j]$$

Plugging in the factorizations for $\hat{Q}[k, j]$ and $\hat{\mathcal{K}}[k, i, j]$, the residual becomes:

$$\mathcal{R}_{qk}[k, i, j] = Q[k, i] - \Phi^+[k, i] \cdot \hat{K}[k, j] = \Phi^+[k, i] \cdot (\hat{Q}[k, i] - \hat{K}[k, j]).$$

The matrix of Mahalanobis distances now becomes:

$$D^2[i, j] = \sum_k \underbrace{\mathcal{K}_P[k, |i - j|]}_{\text{Precision kernel}} \cdot \underbrace{|\Phi^+[k, i]|^2}_{\text{Forward decay}} \cdot \left[\underbrace{|\hat{Q}[k, i]|^2}_{\text{Stationary Query}} + \underbrace{|\hat{K}[k, j]|^2}_{\text{Stationary Key}} - \underbrace{2 \operatorname{Re}(\hat{Q}^*[k, i] \hat{K}[k, j])}_{\text{Stationary Cross-term}} \right]$$

(where $*$ denotes the complex conjugate). The remaining bottleneck is the k -dependence of the precision kernel $\mathcal{K}_P$ in the evaluation of the cross-term:

$$\sum_k \mathcal{K}_P[k, |i - j|] \cdot 2 \operatorname{Re}(\hat{Q}^*[k, i] \hat{K}[k, j]).$$

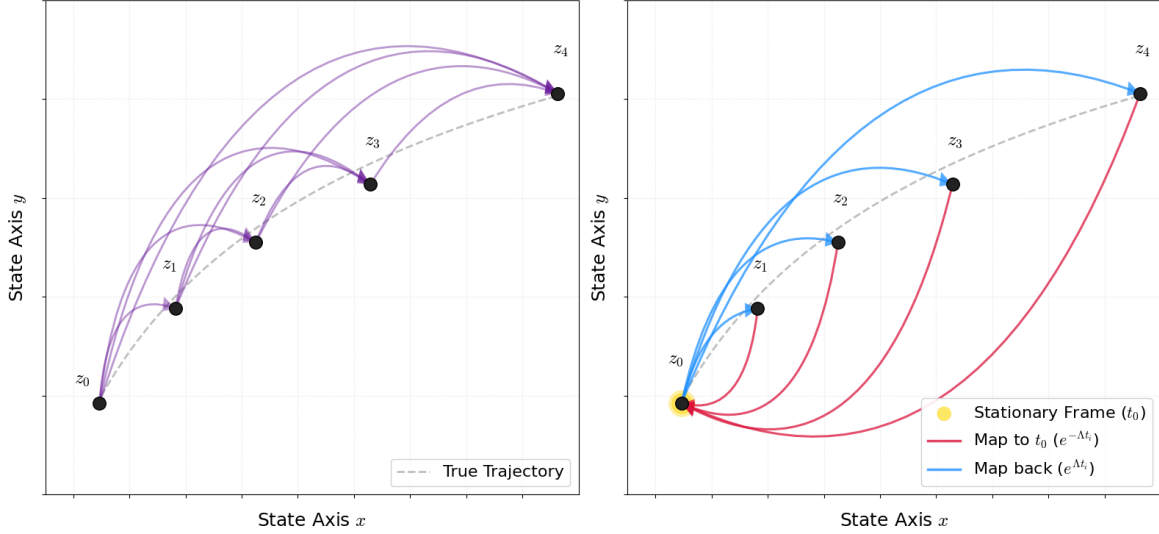


Figure 2. **Factorizing state propagation through a common reference frame.** **Left:** Direct propagation requires a distinct transport operator $e^{\Lambda(t_i-t_j)}$ for every source–destination pair (i, j) . **Right:** Using $e^{\Lambda(t_i-t_j)} = e^{\Lambda t_i} e^{-\Lambda t_j}$, each representation is first transported once into the stationary frame ($t = 0$) via $e^{-\Lambda t_j}$, producing reusable stationary representations. Each query then applies its own forward transport $e^{\Lambda t_i}$ after attention aggregation to recover the output in the query frame. This replaces N^2 pairwise transport operators with two sets of N transport factors.

In standard attention, scores are computed with a single matrix multiplication ($QK^\top$). Here, however, the precision kernel $\mathcal{K}^P[k, |i-j|]$ weights each feature differently as a function of time lag, so the summation over k cannot be expressed as a single matmul. Achieving $\mathcal{O}(N^2 + Nd)$ memory therefore requires the precision kernel to be independent of the feature index k , allowing it to factor outside the summation.

A degenerate case occurs in the zero-noise limit, where the precision kernel is constant. This recovers a memory-efficient formulation with anisotropic (feature-wise) decay, similar to xPos.

However, for stable dynamics with $\mu_k > 0$, the backward transition factor $\Phi^-[k, j] = e^{(\mu_k - i\omega_k)t_j}$ grows exponentially with sequence length. Hence, when decay rates vary across features, the stationary representations $\hat{Q}, \hat{K}, \hat{V}$ grow exponentially with sequence length, making fully parallel computation numerically unstable because forward and backward factors cancel only after multiplication, allowing intermediate values to overflow.

Therefore, retaining a non-constant precision kernel while ensuring numerical stability under extrapolation requires restricting decay to be isotropic within each head. This allows decay to be factored at the head level rather than per feature, enabling stable, fully parallel attention with $\mathcal{O}(N^2 + Nd)$ memory. This motivates the Isotropic RFA variant introduced next.

B.3. Isotropic RFA

B.3.1. ISOTROPIC DECAY AND NOISE ASSUMPTIONS

All assumptions in this section are applied *per attention head*. In particular, the real part of the eigenvalues within a head is taken to be a shared scalar $-\mu$:

$$\lambda_k = -\mu + i\omega_k, \quad \mu \in \mathbb{R}^+, \omega_k \in \mathbb{R}$$

This corresponds to a system with an isotropic plus skew-symmetric state matrix:

$$\mathbf{A} = -\mu \mathbf{I} + \mathbf{\Omega}, \quad \mathbf{\Omega} = -\mathbf{\Omega}^\top \in \mathbb{R}^{d \times d}.$$

We also assume that the noise is isotropic, i.e. that the noise covariances are scalar multiples of identity:

$$\mathbf{\Lambda}_Q = \sigma^2 \mathbf{I}, \quad \mathbf{\Lambda}_R = \eta^2 \mathbf{I}.$$

Under this constraint, the covariance kernel simplifies to a scalar function of the lag $|i-j|$:

$$\Sigma_{\Delta t}[i, j] := \Sigma(|i-j|) = \tilde{\sigma}^2 (1 - e^{-2\mu\delta t|i-j|}) + \eta^2 e^{-2\mu\delta t|i-j|} + \sigma_0^2, \quad \tilde{\Sigma}_{\Delta t}[i, j] = \Sigma_{\Delta t}[i, j] + \gamma^2$$

Hence, the precision kernel becomes a scalar function of the time lag $\tau = |i - j|$, allowing it to be pulled outside the feature summation. Defining $\mathbf{P}_{\Delta t}[i, j] := 1/\Sigma_{\Delta t}[i, j]$ and $\tilde{\mathbf{P}}_{\Delta t}[i, j] := 1/\tilde{\Sigma}_{\Delta t}[i, j]$, the matrix of Mahalanobis distances become:

$$\mathbf{D}^2[i, j] = \tilde{\mathbf{P}}_{\Delta t}[i, j] \cdot \left(\sum_k |\mathcal{R}_{qk}[k, i, j]|^2 \right) =: \tilde{\mathbf{P}}_{\Delta t}[i, j] \cdot \|\mathbf{R}_{qk}[i, j]\|^2.$$

Note that $\|\mathbf{R}_{qk}\|$ denotes a matrix of vector norms, rather than a matrix norm.

B.3.2. SIMPLIFYING THE SQUARED RESIDUAL NORM

The isotropic constraint allows the dynamics to be factored into a stable decay kernel and complex-valued forward/backward rotations:

$$\mathbf{E}[i, j] = e^{-\mu|t_i - t_j|}, \quad \tilde{\Phi}^+[k, i] := e^{i\omega_k t_i}, \quad \tilde{\Phi}^-[k, i] := e^{-i\omega_k t_i},$$

We can then define backward-rotated queries, keys, and values:

$$\tilde{\mathbf{Q}} := \tilde{\Phi}^- \odot \mathbf{Q}, \quad \tilde{\mathbf{K}} := \tilde{\Phi}^- \odot \mathbf{K}, \quad \tilde{\mathbf{V}} := \tilde{\Phi}^- \odot \mathbf{V},$$

Note that:

$$\Phi^+[k, i] = e^{-\mu t_i} \tilde{\Phi}^+[k, i], \quad \hat{\mathbf{Q}}[k, i] = e^{\mu t_i} \tilde{\mathbf{Q}}[k, i], \quad \hat{\mathbf{K}}[k, j] = e^{\mu t_j} \tilde{\mathbf{K}}[k, j].$$

Plugging this into the expression for the Mahalanobis distance, and using the fact that complex rotation preserves magnitude:

$$\begin{aligned} \|\mathbf{R}_{qk}[i, j]\|^2 &= \sum_k e^{-2\mu t_i} |\tilde{\Phi}^+[k, i]|^2 \cdot \left[e^{2\mu t_i} |\tilde{\mathbf{Q}}[k, i]|^2 + e^{2\mu t_j} |\tilde{\mathbf{K}}[k, j]|^2 - 2e^{\mu t_i} e^{\mu t_j} \operatorname{Re}(\tilde{\mathbf{Q}}^*[k, i] \tilde{\mathbf{K}}[k, j]) \right] \\ &= \sum_k \left[|\tilde{\mathbf{Q}}[k, i]|^2 + e^{-2\mu(t_i - t_j)} |\tilde{\mathbf{K}}[k, j]|^2 - 2e^{-\mu(t_i - t_j)} \operatorname{Re}(\tilde{\mathbf{Q}}[k, i]^* \tilde{\mathbf{K}}[k, j]) \right]. \end{aligned}$$

Or, in vectorized form:

$$\|\mathbf{R}_{qk}[i, j]\|^2 = \underbrace{\|\mathbf{Q}_i\|^2}_{\text{Query Norm}} + \underbrace{\mathbf{E}[i, j]^2 \cdot \|\mathbf{K}_j\|^2}_{\text{Decayed Key Norm}} - \underbrace{2 \mathbf{E}[i, j] \cdot \operatorname{Re}(\tilde{\mathbf{Q}}_i^\dagger \tilde{\mathbf{K}}_j)}_{\text{Cross-term}}$$

(since $\|\mathbf{Q}_i\|^2 = \|\tilde{\mathbf{Q}}_i\|^2$ and $\|\mathbf{K}_j\|^2 = \|\tilde{\mathbf{K}}_j\|^2$). Hence, under the isotropic assumption, the cross-term $\operatorname{Re}(\tilde{\mathbf{Q}}^\dagger \tilde{\mathbf{K}})$ can be computed using one $\mathcal{O}(N^2 d)$ matrix multiplication, achieving the required memory efficiency.

B.3.3. THE ATTENTION MATRIX AND ESTIMATE

The logit matrix $\mathbf{L}$ is defined using the power-law/Student- t weighting introduced in Section 3:

$$\mathbf{L} = \log(\mathbf{P}_{\Delta t}) - (\nu_s + 1) \log \left(1 + \frac{1}{\nu_s d} \tilde{\mathbf{P}}_{\Delta t} \odot \|\mathbf{R}_{qk}\|^2 \right),$$

where $\nu_s \in \mathbb{R}^+$ and d is the head dimension.

Defining a causal mask $\mathbf{M}_{\text{causal}}$ and adding an inverse temperature parameter β_s , we can then express the row-normalization using row-wise Softmax:

$$\mathbf{A}[i, j] = \operatorname{Softmax}_j(\beta_s \mathbf{L}[i, j] + \mathbf{M}_{\text{causal}}).$$

The value aggregation is refactored for stability:

$$\begin{aligned} \bar{\mathbf{V}}[k, i] &= \left(e^{-\mu t_i} \tilde{\Phi}^+[k, i] \right) \cdot \sum_{j \leq i} \mathbf{A}[i, j] \cdot \left(e^{\mu t_j} \tilde{\mathbf{V}}[k, j] \right) \\ &= \tilde{\Phi}^+[k, i] \cdot \sum_{j \leq i} \left(\mathbf{A}[i, j] \cdot \mathbf{E}[i, j] \right) \cdot \tilde{\mathbf{V}}[k, j] \end{aligned}$$

Hence, defining a decayed attention matrix $\hat{A} := A \odot E$, the filtered estimate $\bar{V}$ is computed by transforming the aggregation back into the original frame:

$$\bar{V} = \tilde{\Phi}^+ \odot (\tilde{V} \hat{A}^\top).$$

Or, in a form more typical for attention:

$$\bar{V}^\top = (\tilde{\Phi}^+)^\top \odot (\hat{A} \tilde{V}^\top)$$

This rotate–aggregate–counter-rotate structure aligns representations in a shared dynamical frame before aggregation (Fig. 3).

In practice, multi-head attention is implemented by applying this procedure independently across multiple heads, each with its own dynamical and uncertainty parameters. The resulting update steps are concatenated and projected in the standard Transformer manner.

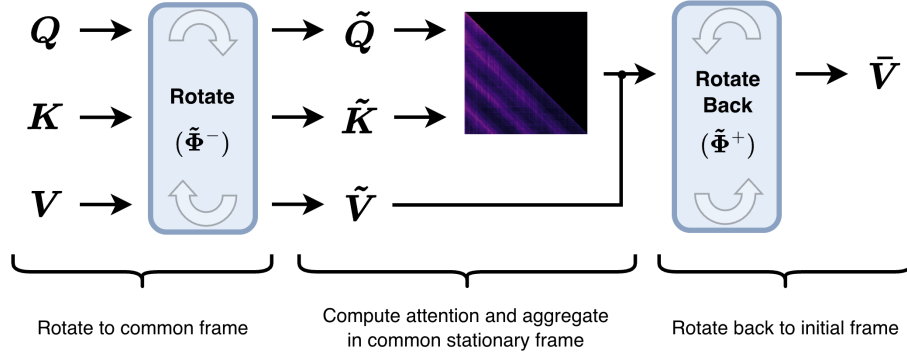


Figure 3. **Rotate, aggregate, counter-rotate structure of Isotropic RFA.** Queries, keys, and values are rotated into a common frame to compute attention and aggregate values. The resulting estimate is then rotated back to the initial frame, yielding a state that preserves relative phase while remaining equivariant to absolute position.

B.3.4. RESIDUAL CONNECTION AND OUTPUT PROJECTION

The filtered value $\bar{V}$ should not be interpreted as a complete replacement for the current representation, but rather as a refined estimate under the robust filtering objective. Accordingly, the attention layer produces a correction relative to the current value representation:

$$\Delta V := \bar{V} - V.$$

The residual connection then applies a partial step toward this estimate:

$$V^+ = V + \alpha \odot \Delta V,$$

where $\alpha \in (0, 1]^d$ is a learned per-feature step size (which may be absorbed into W_o). Mapping back to the original basis:

$$Z^+ = W_o V^+ = W_o V + \alpha \odot W_o \Delta V.$$

Since the value projection maps residual representations into the latent dynamical space while the output projection maps the resulting corrections back into the residual stream, we expect $W_o W_v \approx I$. Under this assumption, the update may be written directly in the residual stream:

$$Z^+ \approx Z + \alpha \odot \Delta Z,$$

where $\Delta Z = W_o \Delta V$. Hence, each attention layer contributes an incremental correction to the current representation.

B.3.5. THE ZERO-DECAY LIMIT AND ALiBi

If the queries and keys are normalized, the squared residual norm becomes

$$\|R_{qk}\|^2 = 1 + E^2 - 2E \odot \text{Re}(\tilde{Q}^\dagger \tilde{K}).$$

In the zero-decay limit ($\mu \rightarrow 0$), the relative decay vanishes ($E = 1$), reducing the residual to the chordal distance on the unit hypersphere:

$$\|R_{qk}\|^2 = 2(1 - \text{Re}(\tilde{Q}^\dagger \tilde{K})).$$

Using the exponential attention kernel gives

$$L = \log(P_{\Delta t}) - \frac{2}{\nu} \tilde{P}_{\Delta t} + \frac{2}{\nu} \tilde{P}_{\Delta t} \odot \text{Re}(\tilde{Q}^\dagger \tilde{K}).$$

In this limit, both $\Sigma_{\Delta t}$ and $\tilde{\Sigma}_{\Delta t}$ grow linearly with temporal lag,

$$\Sigma_{\Delta t} = \sigma^2 \Delta t + \beta, \quad \tilde{\Sigma}_{\Delta t} = \sigma^2 \Delta t + \tilde{\beta},$$

with respective floors $\beta := \eta^2 + \sigma_0^2$ and $\tilde{\beta} := \beta + \gamma^2$, and therefore

$$P_{\Delta t} = \frac{1}{\sigma^2 \Delta t + \beta}, \quad \tilde{P}_{\Delta t} = \frac{1}{\sigma^2 \Delta t + \tilde{\beta}}.$$

The additive positional bias is therefore

$$B_{\Delta t} := \log(P_{\Delta t}) = -\log\left(1 + \frac{\sigma^2}{\beta} \Delta t\right) + \text{const},$$

which, for $\Delta t \ll \beta/\sigma^2$, admits the approximation

$$B_{\Delta t} \approx -\frac{\sigma^2}{\beta} \Delta t + \text{const}.$$

recovering the linear recency bias used by ALiBi, with slope and intercept determined by σ^2 , η^2 , and σ_0^2 .

B.4. Diffusive and Integrative Filtering Regimes

Here we derive the explicit functional forms of the additive bias $B_{\Delta t} := \log(P_{\Delta t})$ and multiplicative gate $\tilde{P}_{\Delta t}$ in each regime, determined by the values of $\alpha := \eta^2 - \tilde{\sigma}^2$, $\beta := \sigma_0^2 + \tilde{\sigma}^2$, and $\tilde{\beta} := \beta + \gamma^2$. The behavior of each regime is illustrated in Fig. 4, and the effect of μ on the speed of the phase transition is shown in Fig. 5.

Diffusive Regime ($\alpha < 0$). Letting $\alpha' = -\alpha > 0$, the bias follows a logarithmic decay:

$$B_{\Delta t} = -\log(\beta) - \log\left(1 - \frac{\alpha'}{\beta} e^{-2\mu \Delta t}\right),$$

starting at its maximum at $\Delta t = 0$ and decaying toward $-\log(\beta)$. The precision decays as:

$$\tilde{P}_{\Delta t} = (\tilde{\beta} - \alpha' e^{-2\mu \Delta t})^{-1},$$

with selectivity maximal near the diagonal and blurring out at longer lags.

Integrative Regime ($\alpha > 0$). The bias follows a mirrored Softplus:

$$B_{\Delta t} = -\log(\beta) - \text{Softplus}(\ln(\alpha/\beta) - 2\mu \Delta t),$$

starting low and curving upward as key-side measurement noise dissipates. The precision follows a sigmoid:

$$\tilde{P}_{\Delta t} = \frac{1}{\tilde{\beta}} \cdot \text{sigmoid}\left(2\mu \Delta t - \ln(\alpha/\tilde{\beta})\right),$$

with selectivity initially low, opening as the transported observation becomes reliable.

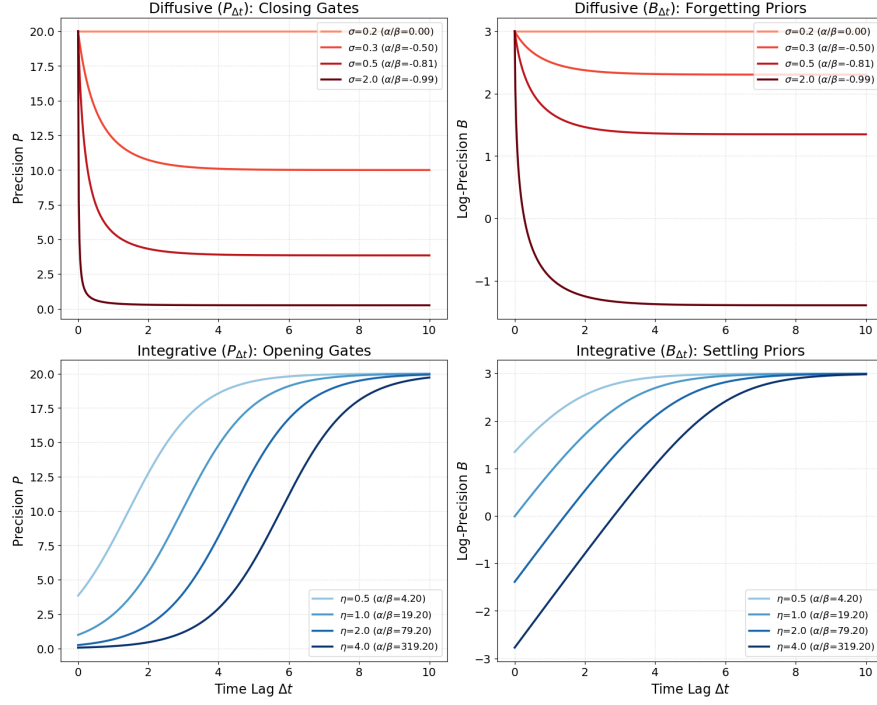


Figure 4. By varying the ratio of steady-state process uncertainty $\tilde{\sigma}^2$ to measurement noise η^2 , RFA heads can specialize into distinct physical regimes: a **diffusive regime** that favors local recency (top row) and an **integrative regime** (bottom row) that filters transient noise to identify stable historical trends. The multiplicative gate $\hat{P}_{\Delta t}$ controls the selectivity (adaptive gain) of the attention, while the additive bias $B_{\Delta t}$ defines the prior budget allocated to tokens at a given temporal lag.

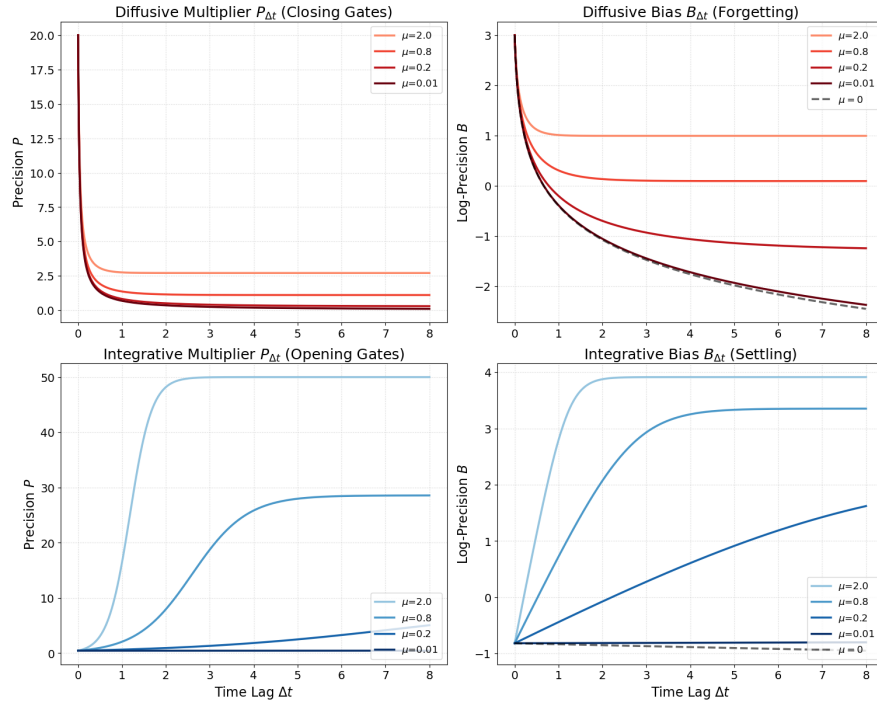


Figure 5. The decay rate μ dictates the speed of the phase transition. As $\mu \rightarrow 0$, the model recovers Brownian diffusion, where precision drops linearly with time. As μ increases, the model enforces stationarity, where the attention bias saturates to a learned global noise floor β , providing a principled mechanism for long-range context retention.

B.5. Spectrally Coupled RFA

In standard RFA, all heads share the full frequency range with a uniform decay rate, so high- and low-frequency components are damped equally within each head (Fig. 6, left). SC-RFA partitions the spectrum across heads and couples each head’s decay rate to its maximum frequency (Fig. 6, right). As illustrated in Fig. 7, this produces an ordered separation of temporal scales: high-frequency heads decay rapidly and act as short-range filters, while low-frequency heads decay more slowly and preserve long-range structure.

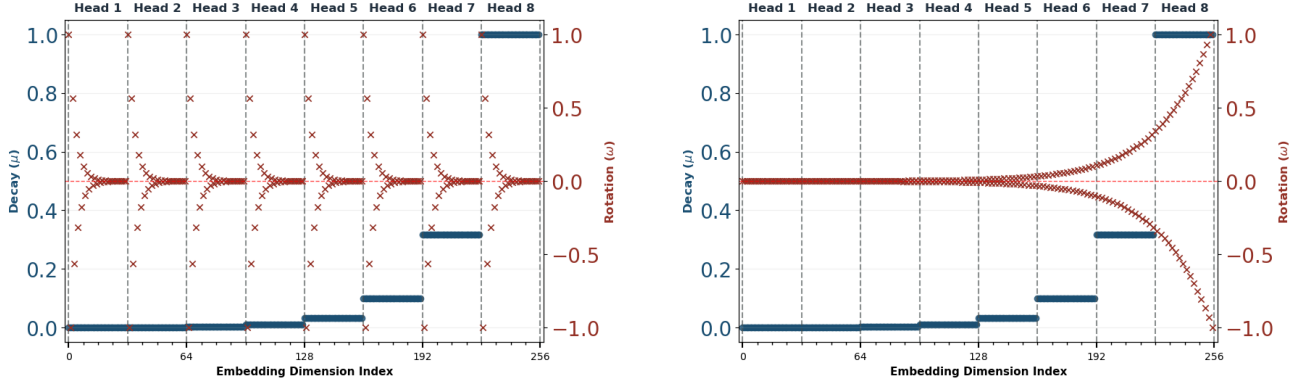


Figure 6. **Eigenvalue spectra of standard isotropic RFA (left) and SC-RFA (right)**, with $b = 1.0$. Standard RFA uses the full frequency range with uniform decay per head. SC-RFA assigns each head a distinct spectral band with decay rate coupled to its maximum frequency. Note that eigenvalues appear in complex-conjugate pairs since the dynamics are real-valued.

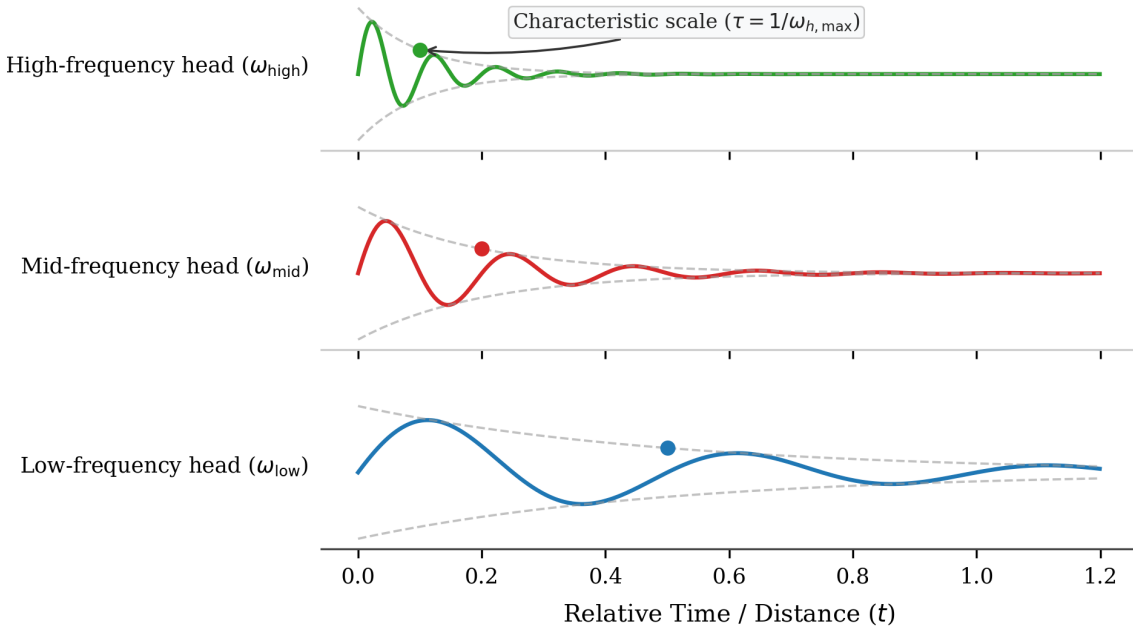


Figure 7. **Temporal scale separation in SC-RFA**. Coupling each head’s decay rate to its maximum frequency ($\mu_h = b\omega_{h,\text{max}}$) causes high-frequency heads to attenuate rapidly while low-frequency heads remain active over longer temporal horizons.

C. Extensions

This section presents some natural generalizations of the RFA framework.

C.1. Distinct Observation Models for Query and Key Streams

The main text assumes a shared observation model $\mathbf{z}_i = \mathbf{C}\mathbf{x}(t_i) + \mathbf{v}_i$. More generally, the query and key streams may arise from distinct observation models:

$$\mathbf{z}_{q,i} = \mathbf{C}_q\mathbf{x}(t_i) + \mathbf{v}_{q,i}, \quad \mathbf{z}_{k,j} = \mathbf{C}_k\mathbf{x}(t_j) + \mathbf{v}_{k,j},$$

where $\mathbf{v}_{q,i} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_q)$, $\mathbf{v}_{k,j} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k)$, and $\mathbf{C}_q, \mathbf{C}_k \in \mathbb{R}^{d \times d}$ are invertible observation operators. Under this interpretation, the query and key streams represent distinct noisy observations of a common latent state. The learned projections $\mathbf{W}_q$ and $\mathbf{W}_k$ can be interpreted as mapping these observations into a shared latent coordinate system, approximately inverting their respective observation operators. In this space, the query–key residual corresponds to a comparison of two views of the same latent state. This provides a probabilistic interpretation for asymmetric query and key projections. The latent state $\mathbf{x}(t)$ evolves under the same SDE as in Section 3.1.

Defining:

$$\mathbf{q}_i := \mathbf{W}_q \mathbf{z}_{q,i}, \quad \mathbf{k}_j := \mathbf{W}_k \mathbf{z}_{k,j}, \quad \mathbf{W}_q := \mathbf{S}^{-1} \mathbf{C}_q^{-1}, \quad \mathbf{W}_k := \mathbf{S}^{-1} \mathbf{C}_k^{-1},$$

the observable residual in the eigenbasis takes the same form as in the main text:

$$\tilde{\mathbf{r}}_{ij} = \mathbf{q}_i - e^{\Lambda \Delta t_{ij}} \mathbf{k}_j.$$

The estimator yields the same precision-weighted form as Eq. 3. Since $\mathbf{W}_k$ maps keys into the latent eigenbasis, the same transformation applies to values, which represent the same latent state estimates being aggregated. Hence, the model implies $\mathbf{W}_v = \mathbf{W}_k$, up to diagonal rescaling.

The residual covariance in the eigenbasis is:

$$\tilde{\Sigma}_{ij} = \mathbf{R}_{q,s} + e^{\Lambda \Delta t_{ij}} \mathbf{R}_{k,s} e^{\Lambda^\dagger \Delta t_{ij}} + \mathbf{V}_{k,s}(\Delta t_{ij}) + \sigma_0^2 \mathbf{I},$$

$$\mathbf{R}_{q,s} = \mathbf{S}^{-1} \mathbf{C}_q^{-1} \mathbf{R}_q \mathbf{C}_q^{-\dagger} \mathbf{S}^{-\dagger}, \quad \mathbf{R}_{k,s} = \mathbf{S}^{-1} \mathbf{C}_k^{-1} \mathbf{R}_k \mathbf{C}_k^{-\dagger} \mathbf{S}^{-\dagger}, \quad \mathbf{V}_{k,s}(\Delta t) = \mathbf{S}^{-1} \mathbf{V}(\Delta t) \mathbf{S}^{-\dagger}.$$

For isotropic measurement noise, $\mathbf{R}_{q,s} = \gamma^2 \mathbf{I}$ and $\mathbf{R}_{k,s} = \eta^2 \mathbf{I}$, $\tilde{\Sigma}_{ij}$ remains diagonal in the eigenbasis and efficient computation of the Mahalanobis distance is preserved.

C.2. Inhomogeneous Dynamics and Measurement Bias

Here, we show that constant drift and measurement bias induce only deterministic shifts in the propagated mean, which can be absorbed into learned bias terms in the query, key, and value projections.

Consider an inhomogeneous linear SDE with constant drift $\mathbf{u}$ and constant measurement bias β :

$$d\mathbf{x}(t) = (\mathbf{A}\mathbf{x}(t) + \mathbf{u}) dt + \mathbf{G} d\mathbf{w}(t), \quad \mathbf{z}_k = \mathbf{C}\mathbf{x}(t_k) + \beta + \mathbf{v}(t_k).$$

The drift term can be eliminated via state augmentation, but this introduces a singular system that breaks simultaneous diagonalization. We therefore treat the inhomogeneous case directly.

We assume $\mathbf{A}$ is Hurwitz. Identifying the equilibrium point μ , the SDE can be rewritten as

$$d\mathbf{x}(t) = \mathbf{A}(\mathbf{x}(t) - \mu) dt + \mathbf{G} d\mathbf{w}(t), \quad \mu = -\mathbf{A}^{-1}\mathbf{u}.$$

The solution to the SDE, propagating the state forward from $\mathbf{x}(t_j)$ to $\mathbf{x}(t_i)$, is

$$\mathbf{x}(t_i) = e^{\mathbf{A} \Delta t_{ij}} \mathbf{x}(t_j) + \left(\int_0^{\Delta t_{ij}} e^{\mathbf{A}(\Delta t_{ij} - \tau)} d\tau \right) \mathbf{u} + \int_0^{\Delta t_{ij}} e^{\mathbf{A}(\Delta t_{ij} - \tau)} \mathbf{G} d\mathbf{w}(\tau).$$

Letting $\mathbf{G}_u(\Delta t_{ij}) = \int_0^{\Delta t_{ij}} e^{-\mathbf{A}\tau} d\tau$, the deterministic part is $\hat{\mathbf{x}}_{ij} = e^{\mathbf{A} \Delta t_{ij}} (\mathbf{x}(t_j) + \mathbf{G}_u(\Delta t_{ij}) \mathbf{u})$. Letting $\mathbf{u}_s := \mathbf{S}^{-1} \mathbf{u}$, the drift term is

$$\mathbf{G}_u(\Delta t_{ij}) \mathbf{u} = \mathbf{S} \left(\int_0^{\Delta t_{ij}} e^{-\Lambda \tau} d\tau \right) \mathbf{u}_s = \mathbf{S} \left(\frac{\mathbf{I} - e^{-\Lambda \Delta t_{ij}}}{\Lambda} \right) \mathbf{u}_s.$$

Hence, the propagated state in the eigenbasis becomes

$$\hat{\mathbf{x}}_{s,ij} = e^{\Lambda \Delta t_{ij}} \left(\hat{\mathbf{x}}_{s,j} + \frac{\mathbf{I} - e^{-\Lambda \Delta t_{ij}}}{\Lambda} \mathbf{u}_s \right) = e^{\Lambda \Delta t_{ij}} \left(\hat{\mathbf{x}}_{s,j} + \frac{\mathbf{u}_s}{\Lambda} \right) - \frac{\mathbf{u}_s}{\Lambda},$$

where division is element-wise. Since the latent state is recovered from the measurements via $\hat{\mathbf{x}}_j = \mathbf{C}^{-1}(\mathbf{z}_j - \boldsymbol{\beta})$, the diagonalized state estimate is

$$\hat{\mathbf{x}}_{s,j} = \mathbf{S}^{-1} \mathbf{C}^{-1} \mathbf{z}_j - \boldsymbol{\beta}_s, \quad \boldsymbol{\beta}_s := \mathbf{S}^{-1} \mathbf{C}^{-1} \boldsymbol{\beta}.$$

Substituting this expression into the propagated state yields

$$\hat{\mathbf{x}}_{s,ij} = e^{\Lambda \Delta t_{ij}} \left(\hat{\mathbf{x}}_{s,j} + \frac{\mathbf{u}_s}{\Lambda} - \boldsymbol{\beta}_s \right) - \left(\frac{\mathbf{u}_s}{\Lambda} - \boldsymbol{\beta}_s \right).$$

Thus, constant drift and measurement bias combine into a single constant offset in the diagonalized coordinates. Because these terms contribute only deterministic shifts, the covariance evolution remains identical to the homogeneous case.

Since this offset is constant across time, it can be absorbed into the learned linear projections by defining bias terms $\mathbf{b}_q, \mathbf{b}_k, \mathbf{b}_v \in \mathbb{C}^{d \times 1}$ in the input projections defining the queries, keys, and values:

$$\mathbf{Q}_u[k, i] := \mathbf{Q}[k, i] + \mathbf{b}_q[k], \quad \mathbf{K}_u[k, i] := \mathbf{K}[k, i] + \mathbf{b}_k[k], \quad \mathbf{V}_u[k, i] := \mathbf{V}[k, i] + \mathbf{b}_v[k],$$

$$\mathbf{b}_\ell[k] := \frac{\mathbf{u}_\ell[k]}{\lambda_{\ell,k}} - \boldsymbol{\beta}_\ell[k], \quad \ell \in \{q, k, v\},$$

where $\mathbf{u}_\ell[k]$, $\boldsymbol{\beta}_\ell[k]$, and $\lambda_{\ell,k}$ denote the k th diagonal element of the projected drift, projected measurement bias, and eigenvalue vector associated with projection ℓ , respectively. These bias terms correspond to the steady-state offset induced by the deterministic dynamics. This allows the residual tensor to maintain the same form as the homogeneous case using the biased tensors:

$$\mathcal{R}_{qk}[k, i, j] = \mathbf{Q}_u[k, i] - \mathcal{E}_{qk}[k, i, j] \cdot \mathbf{K}_u[k, j].$$

The attention output is

$$\bar{\mathbf{V}}[k, i] = \boldsymbol{\Phi}_v[k, i] \cdot \sum_{j \leq i} \mathcal{A}[k, i, j] \cdot \hat{\mathbf{V}}_u[k, j] - \mathbf{b}_v[k] \cdot \sum_{j \leq i} \mathcal{A}[k, i, j],$$

which simplifies, using $\sum_{j \leq i} \mathcal{A}[k, i, j] = 1$, to

$$\bar{\mathbf{V}}[k, i] = \boldsymbol{\Phi}_v[k, i] \cdot \sum_{j \leq i} \mathcal{A}[k, i, j] \cdot \hat{\mathbf{V}}_u[k, j] - \mathbf{b}_v[k].$$

The bias in this final expression can be absorbed into the bias of the output projection, $\mathbf{b}_o := \mathbf{W}_o \mathbf{b}_v$. Hence, the inhomogeneous SDE with constant drift $\mathbf{u}$ and constant measurement bias $\boldsymbol{\beta}$ is structurally equivalent to the homogeneous RFA mechanism, provided these deterministic effects are absorbed into constant bias vectors in the input and output projections ($\mathbf{b}_q, \mathbf{b}_k, \mathbf{b}_v, \mathbf{b}_o$).

C.3. Cross-Attention via Coupled Dynamics

The framework extends naturally to cross-attention by allowing one latent process to influence another. Consider coupled stochastic dynamics:

$$d\mathbf{x}(t) = \mathbf{A}_x \mathbf{x}(t) dt + \mathbf{B} \mathbf{y}(t) dt + \mathbf{G}_x d\mathbf{w}_x(t),$$

$$d\mathbf{y}(t) = \mathbf{A}_y \mathbf{y}(t) dt + \mathbf{G}_y d\mathbf{w}_y(t),$$

$$\mathbf{z}_i^x = \mathbf{C}_x \mathbf{x}(t_i) + \mathbf{v}_i^x, \quad \mathbf{z}_j^y = \mathbf{C}_y \mathbf{y}(t_j) + \mathbf{v}_j^y,$$

where $\mathbf{x}(t)$ denotes the target process and $\mathbf{y}(t)$ an auxiliary process, and where $\mathbf{v}_i^x \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_x)$, $\mathbf{v}_j^y \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_y)$. Applying the variation-of-constants formula yields:

$$\mathbf{x}(t_i) = \boldsymbol{\Phi}_{xx}(i, j) \mathbf{x}(t_j) + \boldsymbol{\Phi}_{xy}(i, j) \mathbf{y}(t_j) + \text{noise},$$

$$\Phi_{xx}(i, j) = e^{\mathbf{A}_x \Delta t_{ij}}, \quad \Phi_{xy}(i, j) = \int_0^{\Delta t_{ij}} e^{\mathbf{A}_x(\Delta t_{ij}-s)} \mathbf{B} e^{\mathbf{A}_y s} ds.$$

Assuming $\mathbf{A}_x$, $\mathbf{A}_y$, and $\mathbf{B}$ share a common eigenbasis,

$$\mathbf{A}_x = \mathbf{S} \mathbf{\Lambda}_x \mathbf{S}^{-1}, \quad \mathbf{A}_y = \mathbf{S} \mathbf{\Lambda}_y \mathbf{S}^{-1}, \quad \mathbf{B} = \mathbf{S} \mathbf{\Lambda}_B \mathbf{S}^{-1},$$

the cross-transport operator remains diagonal in this basis, $\Phi_{xy}(i, j) = \mathbf{S} \mathbf{\Lambda}_{xy}(\Delta t_{ij}) \mathbf{S}^{-1}$. The k -th mode of $\mathbf{\Lambda}_{xy}$ is:

$$\lambda_{xy,k}(\Delta t) = \begin{cases} b_k \frac{e^{\lambda_{y,k} \Delta t} - e^{\lambda_{x,k} \Delta t}}{\lambda_{y,k} - \lambda_{x,k}}, & \lambda_{x,k} \neq \lambda_{y,k}, \\ b_k \Delta t e^{\lambda_{x,k} \Delta t}, & \lambda_{x,k} = \lambda_{y,k}. \end{cases}$$

Defining $\mathbf{q}_i = \mathbf{S}^{-1} \mathbf{C}_x^{-1} \mathbf{z}_i^x$, $\mathbf{k}_j^x = \mathbf{S}^{-1} \mathbf{C}_x^{-1} \mathbf{z}_j^x$, $\mathbf{k}_j^y = \mathbf{S}^{-1} \mathbf{C}_y^{-1} \mathbf{z}_j^y$, the predicted target state in the latent eigenbasis is:

$$\hat{\mathbf{k}}_{ij} = e^{\mathbf{\Lambda}_x \Delta t_{ij}} \mathbf{k}_j^x + \mathbf{\Lambda}_{xy}(\Delta t_{ij}) \mathbf{k}_j^y,$$

and the residual is $\tilde{\mathbf{r}}_{ij} = \mathbf{q}_i - \hat{\mathbf{k}}_{ij}$. For isotropic process and measurement noise, the residual covariance remains scalar and takes the form:

$$\sigma_{\Sigma}^2(i, j) = \sigma_{R_x}^2 + e^{-2\mu_x \Delta t_{ij}} \sigma_{R_x}^2 + |\lambda_{xy}(\Delta t_{ij})|^2 \sigma_{R_y}^2 + \sigma_{V_x}^2(\Delta t_{ij}) + |\lambda_{xy}(\Delta t_{ij})|^2 \sigma_{V_y}^2(\Delta t_{ij}) + \sigma_0^2.$$

Cross-attention is therefore obtained by transporting observations from the auxiliary sequence through Φ_{xy} before computing residuals and precision-weighted aggregation. Unlike standard RFA, the transport kernel is a weighted difference of exponentials, reflecting the interaction of two latent dynamical processes.

C.4. State-Dependent Dynamics

The shared-eigenbasis framework extends naturally to content-dependent dynamics. Let the eigenvalues at token ℓ be:

$$\lambda_{k,\ell} = -\mu_\ell + i\omega_{k,\ell},$$

where $\mu_\ell := \mu(\mathbf{x}_\ell)$, $\omega_{k,\ell} := \omega_k(\mathbf{x}_\ell)$. The dynamics on interval $[\ell, \ell + 1)$ are:

$$\mathbf{A}_\ell = \mathbf{S} \mathbf{\Lambda}_\ell \mathbf{S}^{-1}, \quad \mathbf{\Lambda}_\ell = \text{diag}(\lambda_{1,\ell}, \dots, \lambda_{d,\ell}).$$

Since all intervals share the same eigenbasis, the transition from token j to token i remains diagonalizable:

$$\Phi_{ij} = \prod_{\ell=j}^{i-1} e^{\mathbf{A}_\ell \Delta t} = \mathbf{S} \mathbf{\Gamma}_{ij} \mathbf{S}^{-1}.$$

This can be cheaply computed as a difference of cumulative damping and phase sums:

$$M_i = \sum_{\ell < i} \mu_\ell \Delta t_{\ell, \ell-1}, \quad W_{k,i} = \sum_{\ell < i} \omega_{k,\ell} \Delta t_{\ell, \ell-1},$$

$$\Gamma_{k,ij} = \exp \left(- (M_i - M_j) + i(W_{k,i} - W_{k,j}) \right).$$

The transported observation is $\hat{\mathbf{x}}_{ij} = \Phi_{ij} \hat{\mathbf{x}}_j$. Since only the real part of the eigenvalues contributes to $|\Gamma_{k,ij}|^2$, the content-dependent phase affects transport but not uncertainty propagation. For isotropic process and measurement noise,

$$\lambda_V(i, j) = \sum_{\ell=j}^{i-1} |\Gamma_{k,i,\ell+1}|^2 \lambda_{Q,\ell} \frac{1 - e^{-2\mu_\ell \Delta t}}{2\mu_\ell}.$$

This yields a path-dependent positional encoding in which transport depends on the accumulated content-dependent frequencies encountered between two tokens. Consequently, positional information depends on the sequence content along the path between positions while preserving exact transport and covariance propagation.

C.5. Generalized Analytic Priors via Time-Dependent Noise

The derivation in Section A.1 assumed white process noise. However, each diagonal DLE is a linear ODE, and allowing the noise injection rate $q_k(t)$ to vary in time yields a richer class of analytic priors. For each mode k with decay rate $\mu_k = -\text{Re}(\lambda_k)$, the covariance satisfies:

$$\frac{d}{d\Delta t} \lambda_{V,k}(\Delta t) = -2\mu_k \lambda_{V,k}(\Delta t) + q_k(\Delta t), \quad \lambda_{V,k}(0) = 0.$$

The unique solution is given by the convolution of the mode-specific noise source $q_k(s)$ with the system's exponential impulse response:

$$\lambda_{V,k}(\Delta t) = \int_0^{\Delta t} e^{-2\mu_k(\Delta t-s)} q_k(s) ds.$$

To ensure $\lambda_{V,k}(\Delta t)$ can be solved in closed-form, we restrict the noise source to the class of functions closed under exponential convolution: the complex exponentials. Letting $q_k(s) = \sum_j c_j e^{\gamma_j s}$ for $c_j, \gamma_j \in \mathbb{C}$, the integral yields a weighted sum of exponential differences:

$$\lambda_{V,k}(\Delta t) = \text{Re} \left[\sum_j c_j \left(\frac{e^{\gamma_j \Delta t} - e^{-2\mu_k \Delta t}}{2\mu_k + \gamma_j} \right) \right].$$

This characterizes the most general class of precision kernels that remain analytically tractable under a scalar DLE.

C.6. Stacked Attention Layers as an Unrolled Iterative State Estimator

The robust M-estimator is defined implicitly: the weights w_{ij} depend on residuals computed against the unknown latent state $\mathbf{x}(t_i)$, which must itself be approximated from the current iterate. Each attention layer can therefore be interpreted as one step of an Iteratively Reweighted Least Squares (IRLS)-like procedure: given the previous layer's state estimate, the current layer recomputes residuals, updates weights, and produces a refined precision-weighted average.

Working in the eigenbasis, let $\hat{\mathbf{x}}_{s,i}^{(k)} := \mathbf{S}^{-1} \mathbf{C}^{-1} \tilde{\mathbf{z}}_i^{(k)}$, where the procedure is initialized with each position's own embedding as the zeroth estimate: $\tilde{\mathbf{z}}_i^{(1)} = \mathbf{z}_i$. At each iteration, transported predictions are recomputed from the current state estimates:

$$\hat{\mathbf{x}}_{s,ij}^{(k)} = e^{\Lambda \Delta t_{ij}} \hat{\mathbf{x}}_{s,j}^{(k)}.$$

Weights are recomputed from the Mahalanobis residuals:

$$w_{ij}^{(k)} := \left(1 + (\lambda_{\tilde{P},ij}) |\tilde{\mathbf{r}}_{s,ij}^{(k)}|^2 / \nu \right)^{-\kappa}, \quad \tilde{\mathbf{r}}_{s,ij}^{(k)} := \hat{\mathbf{x}}_{s,i}^{(k)} - \hat{\mathbf{x}}_{s,ij}^{(k)}.$$

A single refinement step k computes the precision-weighted estimate

$$\bar{\mathbf{x}}_{s,i}^{(k)} = \left(\sum_{j \leq i} w_{ij}^{(k)} \lambda_{P,ij} \right)^{-1} \sum_{j \leq i} w_{ij}^{(k)} \lambda_{P,ij} \hat{\mathbf{x}}_{s,ij}^{(k)}.$$

where $\lambda_{P,ij}$ is the precision at lag Δt_{ij} (Appendix A). The estimate may then be updated via an innovation step:

$$\tilde{\mathbf{z}}_i^{(k+1)} = \tilde{\mathbf{z}}_i^{(k)} + \boldsymbol{\alpha}^{(k)} \odot \mathbf{C} \mathbf{S} (\bar{\mathbf{x}}_{s,i}^{(k)} - \hat{\mathbf{x}}_{s,i}^{(k)}),$$

where $\boldsymbol{\alpha}_i \in (0, 1]^d$ controls the correction step size. This mirrors the residual updates used in standard Transformers, which may similarly be interpreted as iterative correction steps. Stacking L attention layers with shared parameters can be interpreted as unrolling L steps of this iterative estimation procedure.

D. Implementation

D.1. Complex-valued Computations

RFA is formulated in a complex latent space. In practice, this is implemented by lifting real-valued representations into a $2d$ -dimensional space using a linear projection (corresponding to $\mathbb{C}^d \cong \mathbb{R}^{2d}$), performing complex rotations and attention in this space, and then projecting the result back to $\mathbb{R}^d$.

A complex-valued linear transformation can be represented in the real domain by operating on paired real and imaginary channels. For an input $\mathbf{x} = [\mathbf{x}_r, \mathbf{x}_i]^\top$ with $\mathbf{x}_r, \mathbf{x}_i \in \mathbb{R}^d$, this corresponds to:

$$\mathcal{L}(\mathbf{x}) = \begin{bmatrix} \mathbf{W}_r & -\mathbf{W}_i \\ \mathbf{W}_i & \mathbf{W}_r \end{bmatrix} \mathbf{x} + \begin{bmatrix} \mathbf{b}_r \\ \mathbf{b}_i \end{bmatrix},$$

Here $\mathbf{W}_r, \mathbf{W}_i \in \mathbb{R}^{d \times d}$ are the real and imaginary components of the weight matrix and $\mathbf{b}_r, \mathbf{b}_i \in \mathbb{R}^d$ the bias. This is equivalent to multiplication by a complex matrix $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i$ with bias $\mathbf{b} = \mathbf{b}_r + i\mathbf{b}_i$.

Assuming the inputs and outputs are purely real, only the real-input columns of the input projections and the real-output columns of the output projections are required:

$$\mathcal{L}^{d \times 2d}(\mathbf{x}_r) := \begin{bmatrix} \mathbf{W}_r \\ \mathbf{W}_i \end{bmatrix} \mathbf{x}_r + \begin{bmatrix} \mathbf{b}_r \\ \mathbf{b}_i \end{bmatrix}.$$

$$\mathcal{L}^{2d \times d}(\mathbf{x}) := [\mathbf{W}_r \quad -\mathbf{W}_i] \mathbf{x} + \mathbf{b}_r.$$

Hence, both projections may be implemented using standard real-valued linear layers in $\mathbb{R}^{2d}$.

We define queries, keys, and values using:

$$\mathbf{Q} = \mathcal{L}_q^{d \times 2d}(\mathbf{Z}), \quad \mathbf{K} = \mathcal{L}_k^{d \times 2d}(\mathbf{Z}), \quad \mathbf{V} = \mathcal{L}_v^{d \times 2d}(\mathbf{Z})$$

We define cosine and sine matrices:

$$\mathbf{C}[k, i] = \cos(\omega_k t_i), \quad \mathbf{S}[k, i] = \sin(\omega_k t_i)$$

Complex rotations are applied as:

$$\tilde{\mathbf{Q}}^\top = \begin{bmatrix} \tilde{\mathbf{Q}}_r^\top \\ \tilde{\mathbf{Q}}_i^\top \end{bmatrix} = (\tilde{\Phi}^-)^\top \odot \mathbf{Q}^\top = \begin{bmatrix} \mathbf{C} \odot \mathbf{Q}_r^\top + \mathbf{S} \odot \mathbf{Q}_i^\top \\ \mathbf{C} \odot \mathbf{Q}_i^\top - \mathbf{S} \odot \mathbf{Q}_r^\top \end{bmatrix},$$

and likewise for $\tilde{\mathbf{K}}$ and $\tilde{\mathbf{V}}$. This is algebraically identical to RoPE.

To ensure the underlying system matrix $\mathbf{A}$ is real-valued, we enforce that its eigenvalues appear in complex conjugate pairs:

$$\omega = \{\omega_1, -\omega_1, \dots, \omega_{d/2}, -\omega_{d/2}\},$$

Hence, only the positive-frequency half of the spectrum is learned explicitly, with the remaining frequencies determined by conjugation. The lifted $2d$ -dimensional representation requires only $d/2$ learned frequencies, rather than the d frequencies that would be required by an unconstrained RoPE parameterization in the same space.

The Mahalanobis distance requires the real part of the complex inner product,

$$\text{Re}(\tilde{\mathbf{Q}}^\dagger \tilde{\mathbf{K}}) = \tilde{\mathbf{Q}}_r^\top \tilde{\mathbf{K}}_r + \tilde{\mathbf{Q}}_i^\top \tilde{\mathbf{K}}_i = \begin{bmatrix} \tilde{\mathbf{Q}}_r^\top & \tilde{\mathbf{Q}}_i^\top \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{K}}_r \\ \tilde{\mathbf{K}}_i \end{bmatrix}.$$

This is implemented as a single real matrix multiplication in $\mathbb{R}^{2d}$.

Value aggregation, $\bar{\mathbf{V}}$, is computed in the $\mathbb{R}^{2d}$ domain. The real-valued attention matrix $\hat{\mathbf{A}}$ is applied identically to both the real and imaginary components of the complex-rotated values:

$$\mathbf{M}^\top = \hat{\mathbf{A}} \begin{bmatrix} \tilde{\mathbf{V}}_r^\top \\ \tilde{\mathbf{V}}_i^\top \end{bmatrix}.$$

The inverse rotation yields:

$$\bar{\mathbf{V}}^\top = (\tilde{\Phi}^+)^\top \odot (\hat{\mathbf{A}} \tilde{\mathbf{V}}^\top) = \begin{bmatrix} \mathbf{C} \odot \mathbf{M}_r^\top - \mathbf{S} \odot \mathbf{M}_i^\top \\ \mathbf{C} \odot \mathbf{M}_i^\top + \mathbf{S} \odot \mathbf{M}_r^\top \end{bmatrix}$$

The value update $\Delta \mathbf{V} = \bar{\mathbf{V}} - \mathbf{V}$ is projected back to the real domain using the $\mathcal{L}^{2d \times d}$ layer:

$$\Delta \mathbf{Z} = \mathcal{L}_o^{2d \times d}(\Delta \mathbf{V}) = [\mathbf{W}_r \quad -\mathbf{W}_i] \Delta \mathbf{V} + \mathbf{b}_r \in \mathbb{R}^{d \times N}.$$

All components of RFA are therefore implemented using standard real-valued operations.

D.2. Initialization

Isotropic Complex Projections. Complex weights $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i$ are initialized isotropically:

$$\mathbf{W}_{ij} = M_{ij} \begin{bmatrix} \cos(\phi_{ij}) \\ \sin(\phi_{ij}) \end{bmatrix}, \quad M_{ij} \sim \text{Rayleigh}\left(\sqrt{\frac{1}{d_{\text{in}} + d_{\text{out}}}}\right), \quad \phi_{ij} \sim \mathcal{U}(0, 2\pi).$$

Output projections ($\mathbf{W}_o$) are scaled by $1/\sqrt{2}$ to preserve variance when converting back to real space.

Noise and Robustness. We define a separate query-side measurement noise γ^2 to give the model extra flexibility. We initialize a constant steady-state uncertainty $\tilde{\sigma}$ across heads by scaling the process noise with the decay rate, $\sigma = 0.1 \mu$. This ensures that the variance floor remains comparable across heads despite differences in temporal persistence. We initialize measurement noise (η^2, γ^2) such that the model begins in the integrative regime ($\eta^2 > \tilde{\sigma}^2$), to preserve long-range gradient flow early in training. We enforce positive query-side measurement noise $\gamma^2 > 0$ to ensure finite precision.

The Student- t degrees of freedom ν are initialized as a positive multiple of the head dimension: $\nu = \nu_s d$. We initialized $\nu_s = 4$, placing the model in a quasi-Gaussian regime during the initial phase of training. This provides a broad prior that prevents the premature rejection of tokens while the Query-Key representations are still unoptimized.

Remark. In our implementation, σ^2 was learned directly. An equivalent and often more numerically stable parameterization is obtained by learning the steady-state variance $\tilde{\sigma}^2 := \sigma^2/(2\mu)$ directly. This decouples the variance floor from the decay rate μ , improving conditioning when μ varies across heads. We drop the update step size α , since it may be absorbed into the output projection $\mathbf{W}_o$.

D.3. Algorithm

Algorithm 1 details the implementation of Isotropic RFA.

(Note: We use $\oplus$ to denote broadcast addition.)

Algorithm 1 Robust Filter Attention (Isotropic; Single Head)

Input: Input sequence $\mathbf{Z} \in \mathbb{R}^{d \times N}$

Definitions:

Linear layers: $\mathcal{L}_q^{d \times 2d}, \mathcal{L}_k^{d \times 2d}, \mathcal{L}_v^{d \times 2d}, \mathcal{L}_o^{2d \times d}$.

Scalar parameters: Noise variance parameters: σ', η', γ' ; robustness parameter ν_s ; Softmax inverse temperature β_s .

Constants: Causal mask $\mathbf{M}_{\text{causal}} \in \{0, -\infty\}^{N \times N}$; angular frequencies ω ; decay rate $\mu \in \mathbb{R}^+$.

Enforce Conjugate Symmetry: $\omega \in \{\omega'_1, -\omega'_1, \dots, \omega'_{d/2}, -\omega'_{d/2}\}$.

Ensure positive noise/decay parameters:

$\{\tilde{\sigma}^2, \eta^2, \gamma^2\} \leftarrow \text{Softplus}(\{\sigma', \eta', \gamma'\})$

Input projections:

$(\text{Re}(\mathbf{Q}), \text{Im}(\mathbf{Q})) \leftarrow \mathcal{L}_q(\mathbf{Z})$

$(\text{Re}(\mathbf{K}), \text{Im}(\mathbf{K})) \leftarrow \mathcal{L}_k(\mathbf{Z})$

$(\text{Re}(\mathbf{V}), \text{Im}(\mathbf{V})) \leftarrow \mathcal{L}_v(\mathbf{Z})$

Decay and rotation kernels: $\mathbf{E}[i, j] = e^{-\mu|t_i - t_j|}$, $\tilde{\Phi}^+[k, i] = e^{i\omega_k t_i}$, $\tilde{\Phi}^-[k, i] = e^{-i\omega_k t_i}$

Covariance kernel: $\Sigma_{\Delta t}[i, j] = \tilde{\sigma}^2(1 - \mathbf{E}[i, j]^2) + \eta^2 \mathbf{E}[i, j]^2 + \sigma_0^2$, $\tilde{\Sigma}_{\Delta t}[i, j] = \Sigma_{\Delta t}[i, j] + \gamma^2$

Query/Key/Value Rotations: $\tilde{\mathbf{Q}}[k, i] = \tilde{\Phi}^- \odot \mathbf{Q}[k, i]$, $\tilde{\mathbf{K}}[k, j] = \tilde{\Phi}^- \odot \mathbf{K}[k, j]$, $\tilde{\mathbf{V}}[k, i] = \tilde{\Phi}^- \odot \mathbf{V}[k, i]$

Squared residuals: $\|\mathbf{R}_{qk}[i, j]\|^2 = \|\mathbf{Q}_i\|^2 + \mathbf{E}[i, j]^2 \cdot \|\mathbf{K}_j\|^2 - 2\mathbf{E}[i, j] \cdot \text{Re}(\tilde{\mathbf{Q}}_i^\dagger \tilde{\mathbf{K}}_j)$

Logits: $\mathbf{L} = -\log(\Sigma_{\Delta t}) - (\nu_s + 1) \log\left(1 + \frac{1}{\nu_s d} \|\mathbf{R}_{qk}\|^2 \odot \tilde{\Sigma}_{\Delta t}\right)$.

Attention matrix: $\mathbf{A}[i, j] = \text{Softmax}_j(\beta_s \mathbf{L}[i, j] + \mathbf{M}_{\text{causal}})$, $\hat{\mathbf{A}} = \mathbf{A} \odot \mathbf{E}$

Value estimate: $\bar{\mathbf{V}} = \tilde{\Phi}^+ \odot (\tilde{\mathbf{V}} \hat{\mathbf{A}}^\top)$

Value step: $\Delta \mathbf{V} = \bar{\mathbf{V}} - \mathbf{V}$

Output projection: $\Delta \mathbf{Z} \leftarrow \mathcal{L}_o(\Delta \mathbf{V})$

Residual connection: $\mathbf{Z}^+ = \mathbf{Z} + \Delta \mathbf{Z}$

Return: $\mathbf{Z}^+$

Our current implementation is written in high-level PyTorch and incurs an approximately $2\times$ training overhead relative to PyTorch's optimized scaled dot-product attention backend; we expect this gap to be reduced with kernel fusion and optimized implementations.

E. Experimental Details and Ablations

E.1. Experimental Setup

Architecture and Model Configuration. All experiments were conducted using a 6-layer decoder-only Transformer architecture. We set the model dimension to $d_{\text{model}} = 256$ with $h = 8$ attention heads. The attention mechanism maps the model dimension to a total latent dimension of 512 via the $d \times 2d$ query, key, and value projections (split into $d_h = 64$ per head), while the $2d \times d$ output projection maps back down to 256.

We employ a Pre-Norm configuration using Layer Normalization. The Feed-Forward Network utilizes an expansion factor of 4. To optimize the parameter budget, we implement weight tying between the token embedding layer and the final linear output head. We use the GPT-2 byte-pair encoding (BPE) tokenizer with a vocabulary size of 50,257 for all language modeling experiments.

To ensure a fair comparison, RFA models and the baselines (RoPE and ALiBi) were designed with near-identical parameter counts. RFA introduces only a small set of scalar coefficients per head to parameterize noise variances ($\tilde{\sigma}^2, \eta^2, \gamma^2$) and robustness (ν, β_s). Hence, the RFA models match the baseline parameter count (19.36M), with only a 0.02% increase due to additional scalar coefficients.

Training and Optimization Protocol. Models were trained for 15 epochs using the Adam optimizer. We utilized a OneCycleLR scheduler with cosine annealing and a 5% warmup period, and trained until convergence. For RFA models, we adopted a decoupled optimization strategy to ensure the stability of the SDE coefficients:

Feature weights use a peak LR of 1×10^{-3} with $\beta_1 = 0.9$, while SDE coefficients use 5×10^{-4} with $\beta_1 = 0.0$ and $\epsilon = 10^{-7}$. We apply global gradient clipping at 1.0, with a stricter 1×10^{-4} threshold for RFA-specific parameters.

All models were trained on the WikiText-103 and BabyLM-2025 datasets using a standard causal language modeling objective.

E.2. Model Variants and Ablation Design

This section presents a series of ablations designed to isolate the contributions of RFA’s core components. The ablations consist of:

1. Baseline models;
2. RFA variants evaluated in Section 4 (M1–M3); and
3. Structural diagnostic ablations (M2.1–M2.6), which progressively remove components of the filtering formulation to test necessity and failure modes. These models are not intended as competitive models, but rather as mechanistic probes of stability and extrapolation behavior.

Baselines:

- **B1: Standard Transformer + RoPE.** Dot-product attention with rotary positional embeddings (Su et al., 2024). Applies $d \rightarrow 2d \rightarrow d$ projections to match RFA parameterization.
- **B2: Standard Transformer + ALiBi.** Dot-product attention with linear distance bias (Press et al., 2022). Tests whether static geometric penalties are sufficient for stability. Applies $d \rightarrow 2d \rightarrow d$ projections to match RFA parameterization.
- **B3: Decayed RoPE.** Identical to B1 but with an additional exponential decay applied to attention scores per-head, as in RFA, testing whether decay alone suffices in the absence of uncertainty modeling.
- **B4: Spectrally Coupled RoPE (SC-RoPE).** Identical to B3 but with frequency-partitioned RoPE with head-wise decay schedules, testing whether decay with spectral coupling can recover SC-RFA’s stability.

Primary RFA Models:

- **M1: Isotropic RFA.** Isotropic RFA as described in Algorithm D.3, replacing the attention module in a standard Transformer. The first two heads are reserved with $\mu_h = 0$, and $\tilde{\sigma}_h$ is initialized to $0.1\mu_h$.
- **M2: Spectrally Coupled RFA (SC-RFA), optimized for near-field performance.** Identical to M1 except with explicit coupling between rotation frequencies and decay rates, $\mu_h = b \cdot \omega_{h,\max}$, using light damping ($b = 0.05$).
- **M3: Spectrally Coupled RFA (SC-RFA), optimized for extrapolation.** Identical to M2 but with stronger damping ($b = 5.0$), and $\tilde{\sigma}_h$ initialized to $0.5\mu_h$.

Structural Diagnostic Ablations: These ablations progressively remove components of the filtering formulation, starting from the full SC-RFA model (M2) and simplifying toward standard attention. Their purpose is to isolate which mechanisms are required for stable extrapolation.

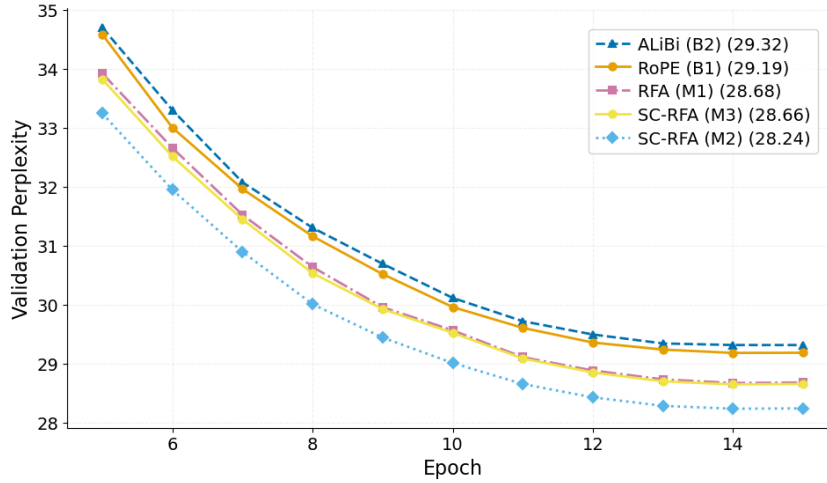
- **M2.1: Exponential Kernel.** M2 with Student’s t influence function replaced by an exponential weighting, i.e., $w_{ij} = \exp(-d_{ij}^2/\nu)$. This isolates the effect of heavy-tailed robust reweighting under the same dynamical precision prior.
- **M2.2: Flat Precision Prior.** M2 with noise parameters removed so that $P_{\Delta t}$ is constant across time lag. Tests whether dynamics alone suffice without uncertainty accumulation.
- **M2.3: No Multiplicative Gate.** M2 with the multiplicative gating term $P_{\Delta t}$ set to a constant, to test the impact of the additive bias $B_{\Delta t}$ in isolation.
- **M2.4: No Value Frame Alignment.** M2 without value rotation and counter-rotation, testing the necessity of aggregating in a shared temporal frame.
- **M2.5: No Rotational Dynamics.** M2 without rotations applied to queries, keys, or values, isolating the effect of decay-only dynamics.
- **M2.6: Unitary Dot-Product Limit.** No decay, no process or measurement noise, and no query and key normalization terms, so that attention weights reduce to normalized complex dot products between rotated embeddings. This yields a purely rotational positional encoding analogous to RoPER (Harik & Jayasiri, 2022).

F. Additional Experimental Results

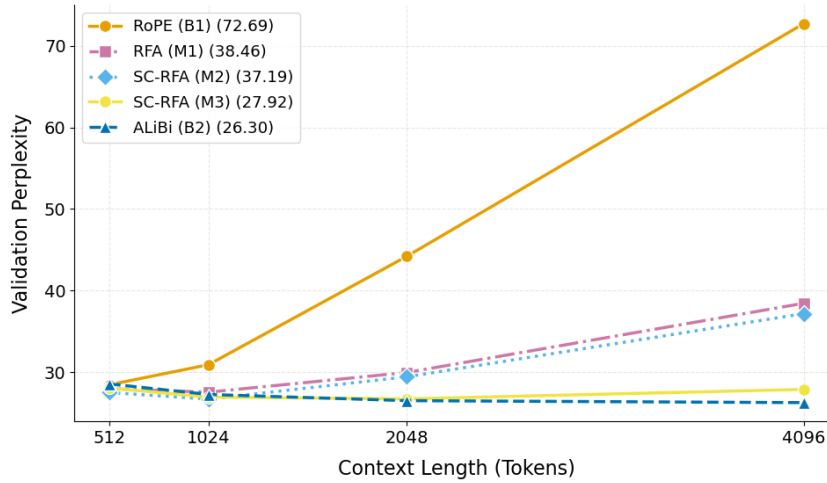
F.1. Training Dynamics and Extrapolation Behavior

To assess learning efficiency, we track validation perplexity throughout training (Fig. 8a). All RFA variants (M1-M3) achieve consistently lower validation perplexity than the baselines (B1 and B2) over the course of training. This suggests that the SDE-based prior provides a more informative inductive bias than purely geometric positional encodings.

Figure 8b visualizes validation PPL and length extrapolation trends corresponding to the tabulated results in Section 4. RFA variants converge faster and degrade more gradually with context length than RoPE, while ALiBi remains stable due to its enforced locality, at the cost of worse performance within the training window.



(a) Validation perplexity over training epochs.



(b) Test perplexity under length extrapolation beyond the training window (512 tokens).

Figure 8. Training dynamics and length extrapolation on WikiText-103. RFA variants converge faster during training and degrade more gradually with increasing context length than RoPE, while ALiBi remains stable due to enforced locality.

Figure 9 shows the sensitivity analysis over damping values b reported in Table 2. Increasing damping improves long-range stability by suppressing high-frequency propagation, but excessively large damping degrades short-range modeling.

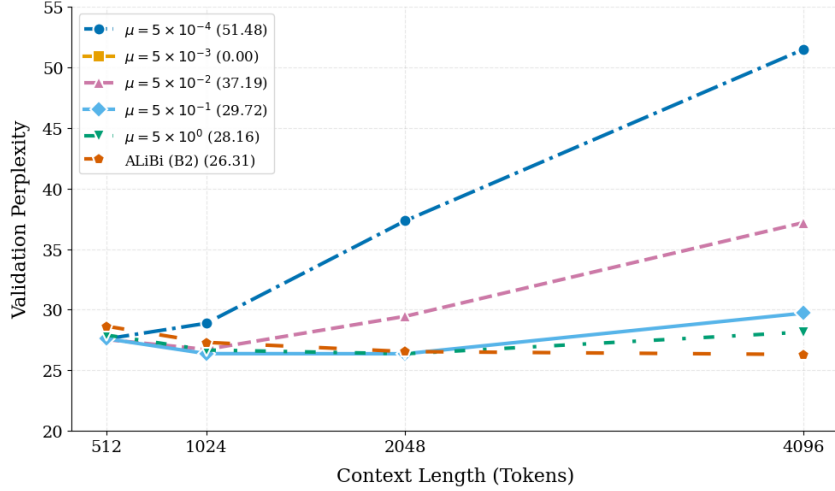


Figure 9. **Impact of the Damping Coefficient b on Length Extrapolation.** Perplexity curves for varying b demonstrate that higher damping coefficient values effectively stabilize long-range integration.

F.2. Parameter Dynamics in RFA

Learned measurement and process noise parameters over the course of training are shown in Fig. 10 for the last layer of both RFA (M1) and SC-RFA (M2). Distinct trajectories in query and key noise parameters indicate that different heads self-organize into separate signal-to-noise regimes. We plot robustness parameter $\nu_s = \nu/d$ and inverse temperature β_s in Fig. 11.

In general, lower-decay heads tend to converge to lower measurement noise variances η^2, γ^2 , lower robustness parameter ν_s , and higher inverse temperature β_s , consistent with stable long-range integration, while higher-decay heads tend to tolerate larger measurement noise.

Intermediate heads tend to converge to the highest measurement noise variance, lowest steady-state process uncertainty, and strongest robustness, consistent with modeling heterogeneous and noisy mid-range structure, while extreme short- and long-range heads tend to remain more tolerant to outliers.

The spectrally coupled model (M2, SC-RFA) exhibits lower average query and key noise variance and more clustered trajectories across heads.

When initialized in the diffusive regime, we observed that higher-decay heads consistently transitioned into the integrative regime ($\alpha > 0$), while the lowest-decay heads remained diffusive (Fig. 12).

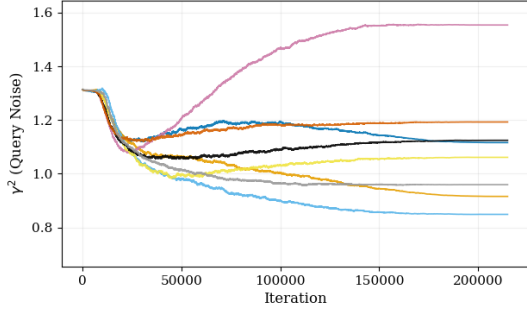
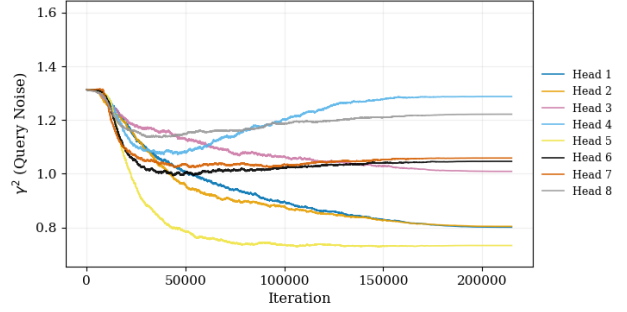
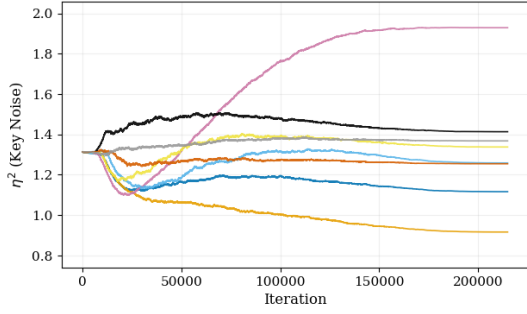
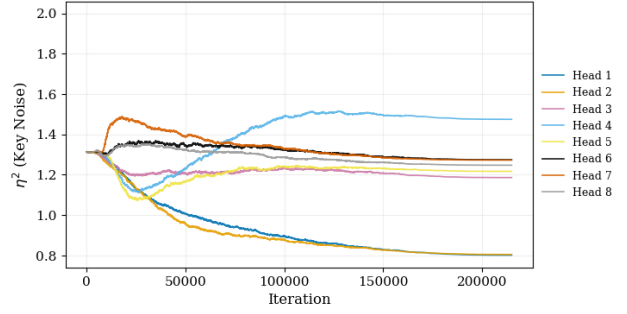
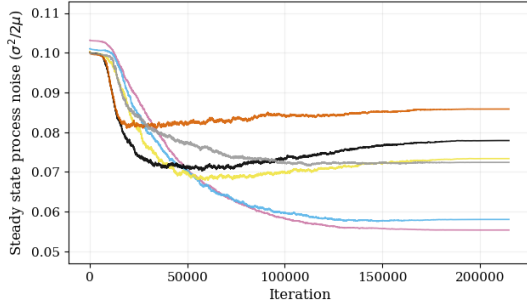
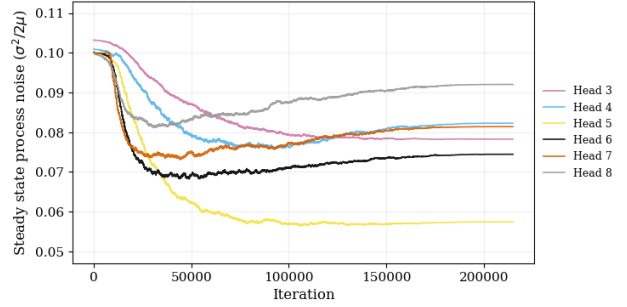

 (a) M1 Query Measurement Noise Variance (γ^2)

 (b) M2 Query Measurement Noise Variance (γ^2)

 (c) M1 Key Measurement Noise Variance (η^2)

 (d) M2 Key Measurement Noise Variance (η^2)

 (e) M1 Steady State Process Variance ($\sigma^2/2\mu$)

 (f) M2 Steady State Process Variance ($\sigma^2/2\mu$)

Figure 10. Measurement and Process Noise Parameters Comparison. Query and key measurement noise variance and state process variance for M1 and M2, over the course of training. (Note that σ^2 is undefined for heads 0 and 1, with $\mu = 0$.)

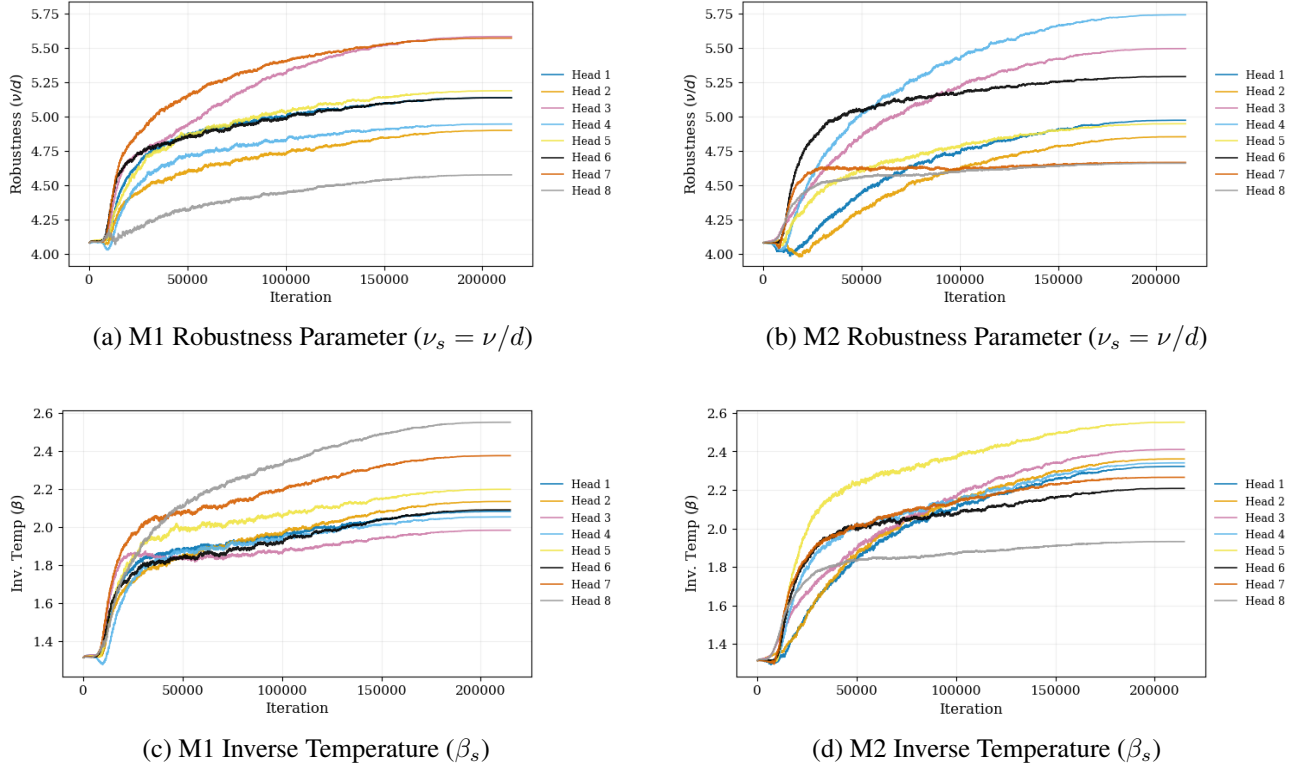


Figure 11. **Robustness and inverse temperature.** Robustness parameter and inverse temperature for M1 and M2, over the course of training.

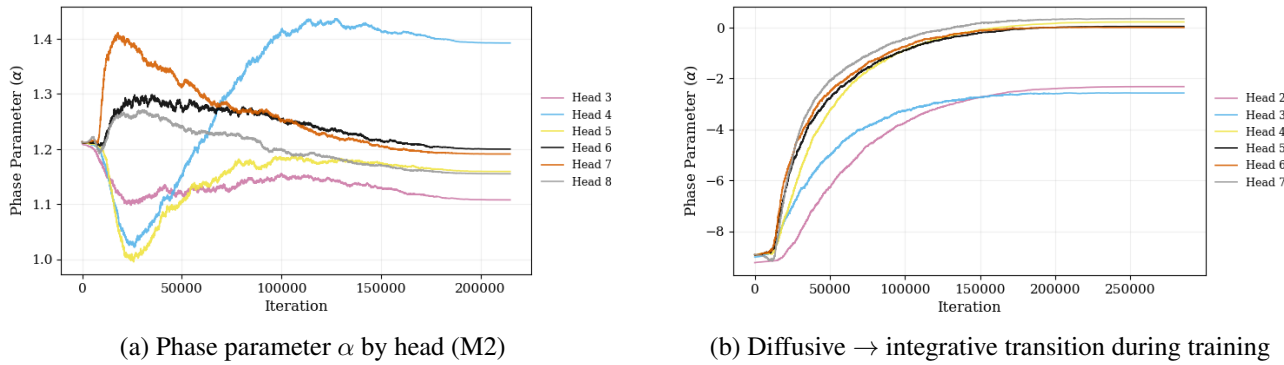


Figure 12. **Integrative dynamics in SC-RFA.** (a) Phase parameter (α) under standard initialization, showing specialization across heads. (b) When initialized in the diffusive regime ($\alpha < 0$), most heads transitioned into the integrative regime ($\alpha > 0$) during training, while the two lowest-decay heads remained diffusive. (Note that α is undefined for heads 0 and 1, with $\mu = 0$.)

E.3. Analysis of Attention Matrices

We plot attention matrices at a context length of 4096 to visualize long-range behaviors induced by each positional prior: the baselines RoPE (B1) (Fig 13) and ALiBi (B2) (Fig 14); and the RFA (M1) (Fig 15) and SC-RFA (M2) (Fig 16) models. We use attention matrices from the last layer of each model.

Note: For the RFA models, for improved visualization, we plot the unattenuated attention matrix $\hat{\mathbf{A}}$ rather than the decayed attention matrix $\hat{\mathbf{A}} := \mathbf{A} \odot \mathbf{E}$.

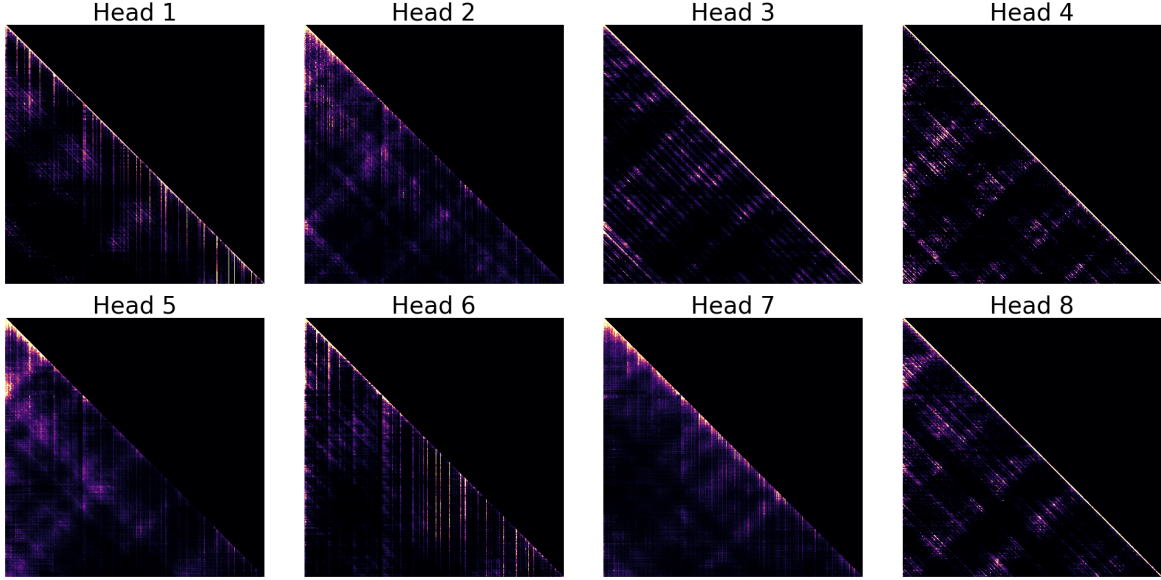


Figure 13. **Baseline RoPE Transformer (B1)** at $L = 4096$: Attention map exhibits persistent checkerboard structure.

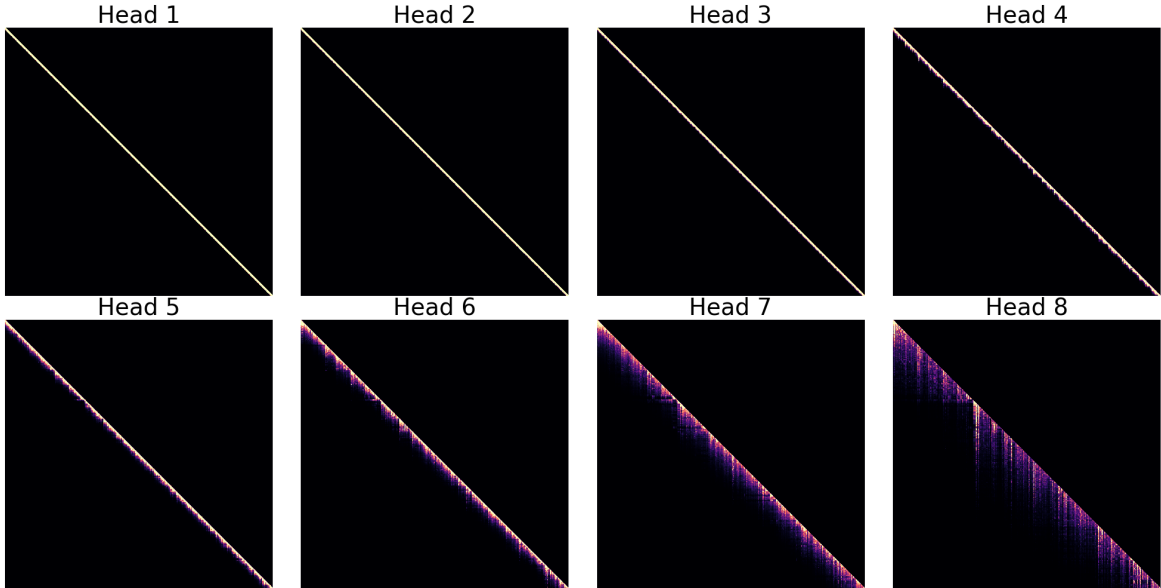


Figure 14. **ALiBi Transformer (B2)** at $L = 4096$: Attention maps remain tightly localized to the diagonal across all heads, with only modest widening in higher heads. Long-range structure is suppressed rather than integrated.

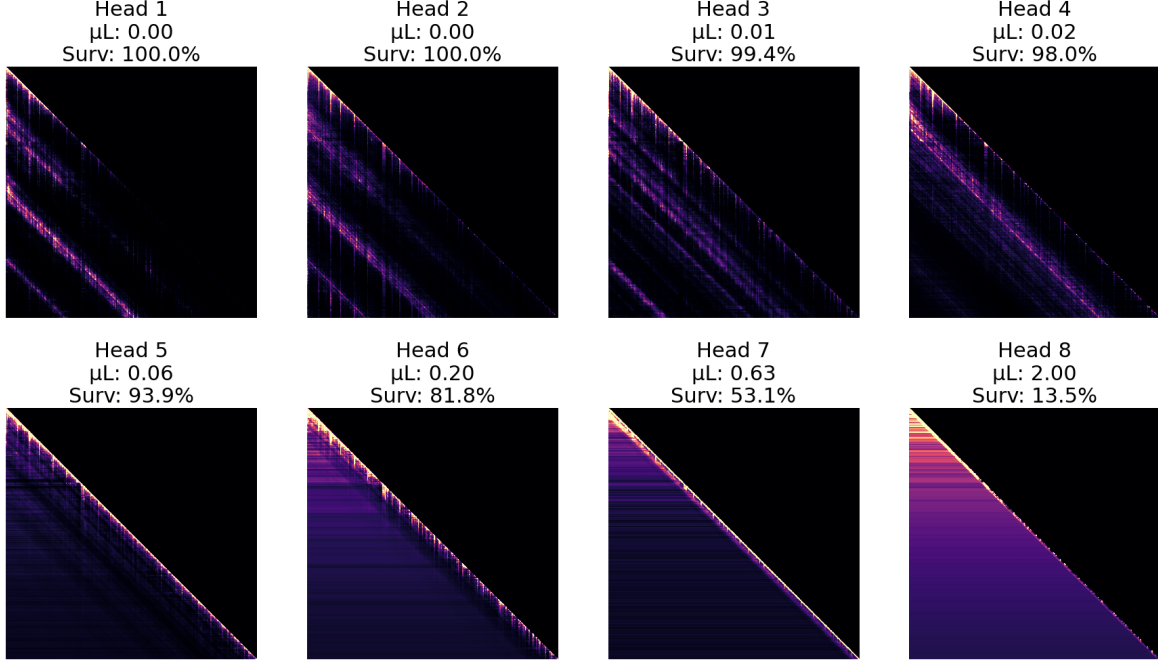


Figure 15. **Robust Filter Attention (M1)** at $L = 4096$: Periodic bands are clearly visible. High-decay heads concentrate focus on the local diagonal, while low-decay heads exhibit the integrative regime: the bottom-right corner near the diagonal is suppressed as the model waits for the SDE dynamics to suppress initial measurement noise before assigning high precision to the state estimate.

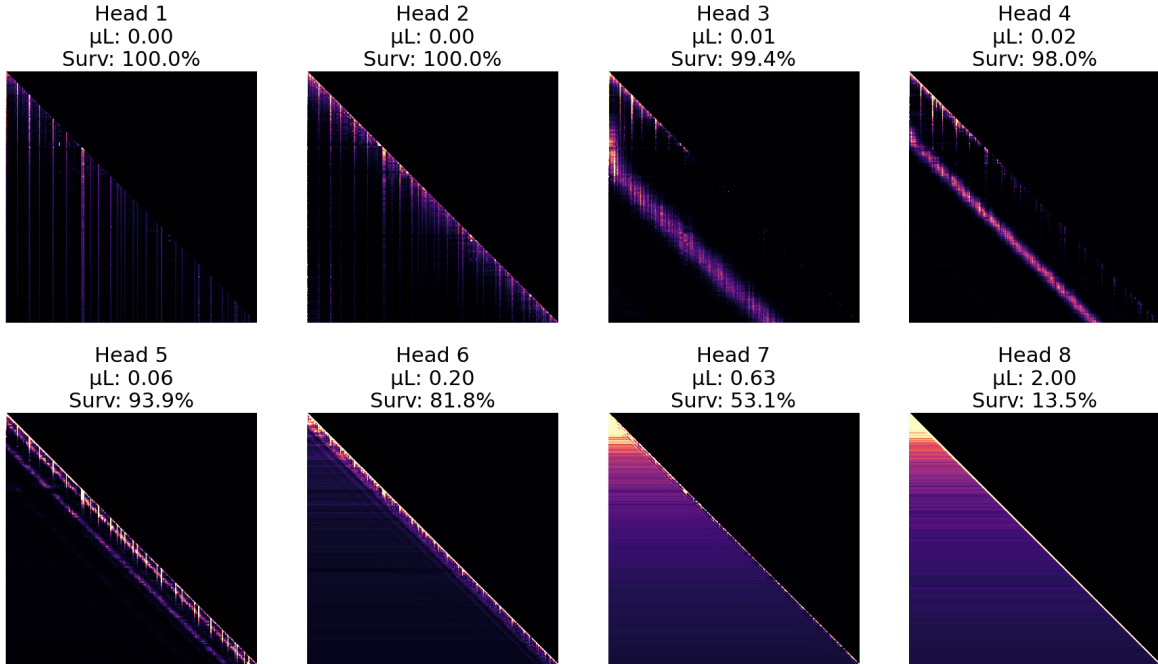


Figure 16. **Spectrally-Coupled RFA (M2, $b = 0.05$)** at $L = 4096$. Frequency-dependent damping ($\mu_h = b \cdot \omega_{h,\max}$) substantially alters long-range attention structure. SC-RFA has fewer periodic bands than RFA. Heads 3-5 each have only a single band, which become narrower and moves closer to the diagonal as decay increases. Heads 1 and 2 act as stable long-range integrators.

The attention maps for RoPE exhibit persistent checkerboard structure and high-frequency oscillations that remain visible even at large temporal offsets. In the absence of decay, these oscillations introduce non-local interference and unstable long-range patterns.

In ALiBi, attention remains tightly localized to the diagonal across all heads, reflecting its fixed distance-based bias. While higher heads show modestly broader receptive fields, long-range context is suppressed rather than integrated.

In contrast, RFA (M1) produces clearer periodic bands by aggregating in a stationary frame, preserving phase relationships and reducing interference. Some heads exhibit an “opening gate” behavior, where attention is suppressed near the diagonal and peaks at a characteristic lag, indicating delayed aggregation until the state estimate stabilizes.

Spectral coupling in SC-RFA (M2) sharpens and organizes this structure. Coupling decay to frequency ($\mu_h = b \cdot \omega_{h,\max}$) induces a specialization of heads to temporal lags: high-decay heads concentrate near the diagonal, while low-decay heads shift toward longer lags, forming distinct, narrow bands consistent with $\Delta t^* \propto 1/\mu_h$ (Sec. 3.7).

In SC-RFA, the first two (zero-decay) heads exhibit vertical structures corresponding to stable long-range retrieval. These heads effectively perform global key–value lookup rather than temporal filtering, assigning similar weight to salient tokens across all query positions and producing clean, vertically aligned patterns.

Together, these patterns support the view that RFA learns a structured multi-scale filtering behavior rather than relying on fixed geometric positional biases.